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**PERIODIC SOLUTIONS FOR NONLINEAR DIFFERENTIAL EQUATIONS IN BANACH  
SPACES WITH GENERALIZED VARIABLE-EXPONENT OPERATORS**

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MEMORIA PARA OPTAR AL TÍTULO DE INGENIERO CIVIL MATEMÁTICO

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## PERIODIC SOLUTIONS FOR NONLINEAR DIFFERENTIAL EQUATIONS IN BANACH SPACES WITH GENERALIZED VARIABLE-EXPONENT OPERATORS

This thesis studies periodic differential equations in Banach spaces of the form

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases}$$

where  $X$  is a real separable reflexive Banach space and  $S : [0, T] \times X \rightarrow X^*$  is a nonlinear time-dependent monotone operator. The framework is motivated by generalized variable-exponent operators and also covers  $p$ -Laplacian-type,  $\phi$ -Laplacian-type, and related monotone laws.

Periodic problems driven by nonlinear operators of this kind have been widely studied in finite-dimensional spaces, particularly in  $\mathbb{R}^N$ , through topological methods. The Banach space setting considered here extends this perspective to broader models, including equations related to nonlinear PDEs, but also introduces new difficulties, since compactness is no longer necessarily automatic.

The analysis starts from the structural properties of  $S$ . These properties allow us to solve the associated non-homogeneous problem and rewrite the periodic equation as a fixed point equation coupled with an averaged constraint. This reformulation leads naturally to a generalized Lyapunov–Schmidt system.

From this formulation, we develop two complementary families of existence results. The first one is based on a class of nonlinearities which we call *mean-admissible*. For this class, the averaged constraint can be solved with respect to the constant component of the solution, reducing the problem to a fixed point equation. The compactness required for the Leray–Schauder degree is recovered through a compact Gelfand triple.

The second family is based on a Hartman-type condition for nonlinearities of the form  $f = m + k$ , where  $m$  is monotone and  $k$  is sequentially weak-to-strong continuous. Existence is obtained through Galerkin approximations, topological degree, and a Minty argument. The purely monotone case is recovered by taking  $k \equiv 0$ , whereas the purely weak-to-strong case is degenerate in infinite-dimensional spaces under the exterior Hartman condition.

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## **SOLUCIONES PERIÓDICAS PARA ECUACIONES DIFERENCIALES NO LINEALES EN ESPACIOS DE BANACH CON OPERADORES TIPO EXPONENTE VARIABLE GENERALIZADO**

Esta tesis estudia ecuaciones diferenciales periódicas en espacios de Banach de la forma

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases}$$

donde  $X$  es un espacio de Banach real, separable y reflexivo, y  $S : [0, T] \times X \rightarrow X^*$  es un operador no lineal monótono dependiente del tiempo. El marco está motivado por operadores generalizados de exponente variable y también incluye operadores de tipo  $p$ -Laplaciano y de tipo  $\phi$ -Laplaciano.

Problemas periódicos de este tipo han sido ampliamente estudiados en  $\mathbb{R}^N$  mediante métodos topológicos. El marco en espacios de Banach extiende esta perspectiva a modelos más amplios, incluyendo ecuaciones relacionadas con EDP no lineales, aunque plantea dificultades adicionales por la falta de compacidad.

El análisis parte de las propiedades estructurales de  $S$ . Estas permiten resolver el problema no homogéneo asociado y reescribir la ecuación periódica como una ecuación de punto fijo acoplada con una restricción promediada, conduciendo a un sistema generalizado de Lyapunov–Schmidt.

A partir de esta formulación, desarrollamos dos familias complementarias de resultados de existencia. La primera se basa en las no linealidades que denominamos *mean-admissible*. Para esta clase, la restricción promediada puede resolverse respecto de la componente constante de la solución, reduciendo el problema a una ecuación de punto fijo. La compacidad necesaria para aplicar el grado de Leray–Schauder se recupera mediante un triple de Gelfand compacto.

La segunda familia se basa en una condición de tipo Hartman para no linealidades de la forma  $f = m + k$ , donde  $m$  es monótona y  $k$  es secuencialmente débil-fuerte continua. La existencia se obtiene mediante aproximaciones de Galerkin, grado topológico y un argumento de Minty. El caso monótono se recupera tomando  $k \equiv 0$ , mientras que el caso puramente débil-fuerte resulta degenerado en dimensión infinita bajo la condición exterior de Hartman.

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# Table of Content

|                                                                                   |           |
|-----------------------------------------------------------------------------------|-----------|
| <b>1. Introduction</b>                                                            | <b>1</b>  |
| <b>2. Preliminaries</b>                                                           | <b>4</b>  |
| 2.1. Topological degree . . . . .                                                 | 4         |
| 2.2. Carathéodory functions and Nemytskii operators . . . . .                     | 5         |
| 2.3. Bochner and variable-exponent spaces . . . . .                               | 6         |
| 2.4. Monotone operators . . . . .                                                 | 9         |
| <b>3. The main problem and its abstract formulation</b>                           | <b>10</b> |
| 3.1. Structural assumptions on the operator $S$ . . . . .                         | 11        |
| 3.2. Solving an implicit equation . . . . .                                       | 13        |
| 3.3. The generalized Lyapunov–Schmidt system . . . . .                            | 18        |
| <b>4. Existence for <i>mean-admissible</i> nonlinearities</b>                     | <b>22</b> |
| 4.1. The class of <i>mean-admissible</i> operators . . . . .                      | 22        |
| 4.2. Recovering compactness through a Gelfand triple . . . . .                    | 28        |
| 4.3. A continuation theorem . . . . .                                             | 31        |
| 4.4. Application to a periodic problem with a sign-changing linear term . . . . . | 34        |
| <b>5. A Hartman-type existence result</b>                                         | <b>38</b> |
| 5.1. Additional assumptions and functional setting . . . . .                      | 39        |
| 5.2. The main Hartman-type theorem . . . . .                                      | 41        |
| 5.3. The limiting cases . . . . .                                                 | 52        |
| 5.3.1. The purely monotone case . . . . .                                         | 52        |
| 5.3.2. The purely weak-to-strong case . . . . .                                   | 53        |
| 5.4. The radial Hilbert-space case . . . . .                                      | 54        |
| 5.5. Application to a spatially nonlocal periodic PDE . . . . .                   | 57        |
| <b>6. Conclusion</b>                                                              | <b>60</b> |
| <b>Bibliography</b>                                                               | <b>62</b> |

# Declaración de uso de inteligencia artificial

En la elaboración de esta tesis se utilizó ChatGPT, herramienta de inteligencia artificial generativa desarrollada por OpenAI, como apoyo en distintas etapas del trabajo. Su uso se concentró exclusivamente en la revisión y mejora de la redacción, la corrección gramatical y de estilo, y la revisión de código en  $\text{\LaTeX}$ .

Todo el contenido incorporado en esta tesis fue revisado, modificado y validado críticamente por el autor. La formulación de los problemas, la selección de las hipótesis, el desarrollo de los resultados, la elaboración de las demostraciones, la interpretación matemática y las conclusiones corresponden al autor, quien asume plena responsabilidad académica por la totalidad del trabajo presentado.



# Chapter 1

## Introduction

Many nonlinear phenomena are governed by constitutive laws whose response is not linear and does not necessarily follow a fixed power law. This occurs, for instance, in models of non-Newtonian fluids, nonlinear elasticity, heterogeneous media, and diffusion processes where the behavior of the system may depend on the position, on time, or on the magnitude of the gradient. From the mathematical point of view, such models lead naturally to differential equations driven by nonlinear operators with nonstandard growth.

Typical examples are given by equations involving  $p$ -Laplacian,  $\phi$ -Laplacian, and related nonlinear operators. More general situations arise when the growth is allowed to vary, as in variable-exponent models, or when the principal operator depends explicitly on time. The functional framework behind these problems is closely related to variable-exponent spaces and related function spaces; see, for instance, [5, 10, 20, 21].

Problems of this type have been studied both for ordinary differential equations and for partial differential equations. In the finite-dimensional setting, several classes of periodic and boundary value problems involving  $p$ -Laplacian-like operators,  $\phi$ -Laplacian-type operators, and generalized variable-exponent operators have been treated by topological methods; see [8, 9, 11, 12]. In the context of partial differential equations, we also mention the works [24–27], where boundary value problems involving weighted  $p(r)$ -Laplacian systems and variable-exponent mean curvature operators are considered.

A direct motivation for this thesis comes from the finite-dimensional result obtained by the author jointly with Raúl Manásevich in [13]. In that work, parameter-dependent periodic systems of ordinary differential equations driven by generalized variable-exponent operators were studied through a fixed point reformulation, Leray–Schauder degree, and global continuation arguments. The results obtained there show that these methods are effective in finite dimension, where compactness is naturally available.

Periodic problems for nonlinear second order differential equations have also been studied in

infinite-dimensional settings, particularly in Hilbert spaces; see, for example, [14, 16, 17].

Motivated by these works and by the finite-dimensional theory described above, one of the aims of this thesis is to investigate how far such methods can be extended to a Banach-space setting. In this context, two difficulties become central: compactness is no longer automatic, and the principal part is not a linear Fredholm operator. These features require a different functional-analytic framework.

We consider periodic differential equations in Banach spaces of the form

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad (1.1)$$

where  $X$  is a real separable reflexive Banach space,  $S : [0, T] \times X \rightarrow X^*$  is an operator which may depend explicitly on time, and  $f : [0, T] \times X \rightarrow X^*$  is a suitable nonlinearity.

The assumptions imposed on  $S$  include continuity, monotonicity, and coercivity properties, while allowing for nonstandard growth. In particular, the abstract setting is designed to include generalized variable-exponent operators as a motivating class, as well as operators related to  $p$ -Laplacian,  $\phi$ -Laplacian, and double-phase type structures.

The dependence of  $S$  on time and the periodic boundary conditions introduce several difficulties. In particular, the associated non-homogeneous problem requires a careful treatment of the periodicity condition. The first step of our analysis is therefore to construct an abstract nonlinear Green operator and use it to obtain an equivalent formulation of (1.1).

More precisely, writing  $u = a + v$ , where  $a \in X$  and  $v \in C([0, T]; X)$  satisfies  $v(0) = v(T) = 0$ , the periodic problem can be reformulated as a generalized Lyapunov–Schmidt system of the form

$$\begin{cases} v = \mathcal{T}(N_f(a + v)), \\ QN_f(a + v) = 0, \end{cases} \quad (1.2)$$

where  $\mathcal{T}$  denotes the nonlinear Green operator associated with the non-homogeneous problem,  $N_f$  is the Nemytskii operator associated with  $f$ , and

$$Qh := \int_0^T h(t) dt.$$

The second equation in (1.2) is the integral constraint induced by the periodic boundary conditions.

The main purpose of this thesis is to establish existence results for the periodic problem (1.1). To this end, we develop complementary methods based on fixed point theory, monotone operators, and finite-dimensional approximations.

The first approach is based on solving the integral constraint with respect to the component  $a$ . For this purpose, we introduce the class of *mean-admissible* nonlinearities. For a nonlinearity in

this class, the constraint can be solved uniquely in terms of  $v$ , reducing the generalized Lyapunov–Schmidt system to a fixed point equation.

In infinite-dimensional spaces, the compactness required to apply the Leray–Schauder degree is not automatic. To recover this property, we work within a compact Gelfand triple

$$X \hookrightarrow H \hookrightarrow X^*,$$

where  $H$  is a real Hilbert space and the embedding  $X \hookrightarrow H$  is compact and dense. This structure provides the compactness needed for a continuation argument. As a consequence, we obtain an existence theorem for mean-admissible nonlinearities under suitable a priori bounds.

The second approach considers nonlinearities of the form

$$f(t, x) = m(t, x) + k(t, x),$$

where  $m(t, \cdot)$  is monotone and  $k(t, \cdot)$  is sequentially weak-to-strong continuous. The generalized Lyapunov–Schmidt system is treated through Galerkin approximations, topological degree, and a Minty argument under an exterior Hartman condition.

The purely monotone case is recovered by taking  $k \equiv 0$ . In this case, the Hartman condition need only be imposed on a sphere and additional uniqueness properties follow. The purely weak-to-strong case is shown to be degenerate in infinite-dimensional spaces. These results extend the finite-dimensional Hartman-type principles in [7, 8] to differential equations in reflexive Banach spaces. For radial operators in Hilbert spaces, the Hartman approach also provides the corresponding pointwise bound.

To the best of our knowledge, the main results of this thesis are new. In particular, they extend the finite-dimensional framework of [8] and a related Hilbert-space result of Mawhin and Willem [17] to reflexive Banach spaces.

The thesis is organized as follows. In Chapter 2, we collect the auxiliary results from topological degree theory, Bochner and variable-exponent spaces, Nemytskii operators, and monotone operator theory that will be used throughout the thesis.

In Chapter 3, we introduce the main problem, construct the nonlinear Green operator, and derive a generalized Lyapunov–Schmidt system associated with the periodic problem.

In Chapter 4, we introduce the class of *mean-admissible* nonlinearities, recover compactness through a Gelfand triple, and prove a continuation theorem based on the Leray–Schauder degree.

In Chapter 5, we prove a Hartman-type existence theorem for nonlinearities of the form  $f = m + k$ , discuss the purely monotone and purely weak-to-strong cases, and establish a pointwise bound for radial operators in Hilbert spaces. Finally, we summarize the main results and discuss possible directions for future research.

# Chapter 2

## Preliminaries

This chapter recalls the analytic tools used throughout the thesis. The aim is not to give a comprehensive account of these topics, but only to collect the definitions and properties needed later. We review topological degree, Nemytskii operators, Bochner and variable-exponent spaces, and monotone operator theory. For general background on nonlinear functional analysis, Banach spaces, compactness, and monotone methods, we refer to [1–4].

### 2.1. Topological degree

We shall use both the Brouwer degree in finite-dimensional spaces and the Leray–Schauder degree in Banach spaces. The presentation below only recalls the properties needed later. For a more detailed treatment of Brouwer degree, Leray–Schauder degree, and continuation methods, see [4, 6, 19, 22, 23]. We also refer to [15] for related Leray–Schauder continuation results.

Let  $\Omega \subset \mathbb{R}^N$  be a bounded open set and let  $f : \overline{\Omega} \rightarrow \mathbb{R}^N$  be continuous. If  $p \notin f(\partial\Omega)$ , the Brouwer degree of  $f$  in  $\Omega$  with respect to  $p$  is denoted by

$$d_B[f, \Omega, p].$$

Let  $E$  be a Banach space and let  $\Omega \subset E$  be a bounded open set. An operator  $T : \overline{\Omega} \rightarrow E$  is called completely continuous if it is continuous and maps bounded sets into relatively compact sets. If  $T : \overline{\Omega} \rightarrow E$  is completely continuous and  $p \notin (I - T)(\partial\Omega)$ , the Leray–Schauder degree of  $I - T$  in  $\Omega$  with respect to  $p$  is denoted by

$$d_{LS}[I - T, \Omega, p].$$

We shall use the following standard properties.

**Proposition 2.1.1** *The Brouwer and Leray–Schauder degrees satisfy the following properties.*

1. **Existence.** *If  $d_B[f, \Omega, p] \neq 0$ , then there exists  $x \in \Omega$  such that  $f(x) = p$ . Similarly, if  $d_{LS}[I - T, \Omega, p] \neq 0$ , then there exists  $x \in \Omega$  such that  $x - T(x) = p$ .*
2. **Homotopy invariance.** *Let  $H : [0, 1] \times \overline{\Omega} \rightarrow \mathbb{R}^N$  be continuous. If  $p \notin H(\lambda, \partial\Omega)$  for every  $\lambda \in [0, 1]$ , then  $d_B[H(\lambda, \cdot), \Omega, p]$  is independent of  $\lambda$ .  
Likewise, if  $\mathcal{H} : [0, 1] \times \overline{\Omega} \rightarrow E$  is completely continuous and*

$$p \notin (I - \mathcal{H}(\lambda, \cdot))(\partial\Omega) \quad \text{for every } \lambda \in [0, 1],$$

*then*

$$d_{LS}[I - \mathcal{H}(\lambda, \cdot), \Omega, p]$$

*is independent of  $\lambda$ .*

3. **Normalization.** *If  $p \in \Omega$ , then  $d_B[I, \Omega, p] = 1$  in finite dimension and  $d_{LS}[I, \Omega, p] = 1$  in Banach spaces.*
4. **Reduction to finite dimension.** *Let  $E_N \subset E$  be a finite-dimensional subspace. If  $p \in E_N$  and  $T(\overline{\Omega}) \subset E_N$ , then*

$$d_{LS}[I - T, \Omega, p] = d_B[(I - T)|_{E_N}, \Omega \cap E_N, p],$$

*whenever both sides are well defined.*

## 2.2. Carathéodory functions and Nemytskii operators

We recall the basic measurability and continuity facts about Nemytskii operators that will be used below. For background on measurable mappings and Nemytskii operators in Banach spaces, see, for instance, [1, 4].

**Definition 2.2.1** (Carathéodory function) Let  $(S, \Sigma)$  be a measurable space, let  $X$  be a topological space, and let  $Y$  be a metrizable space. A function  $f : S \times X \rightarrow Y$  is called a *Carathéodory function* if  $s \mapsto f(s, x)$  is measurable for every  $x \in X$ , and  $x \mapsto f(s, x)$  is continuous for a.e.  $s \in S$ .

**Proposition 2.2.2** Let  $X$  and  $Y$  be separable Banach spaces, and let  $g : [0, T] \times X \rightarrow Y$  be a Carathéodory function. Assume that for every  $\rho > 0$  there exists  $\alpha_\rho \in L^1(0, T)$  such that, for a.e.  $t \in (0, T)$  and every  $x \in X$  with  $\|x\|_X \leq \rho$ ,

$$\|g(t, x)\|_Y \leq \alpha_\rho(t).$$

Then the Nemytskii operator

$$N_g : C([0, T]; X) \rightarrow L^1([0, T]; Y), \quad N_g(u)(t) := g(t, u(t)),$$

is continuous and maps bounded sets into equi-integrable sets.

PROOF. Let  $u_n \rightarrow u$  in  $C([0, T]; X)$ . Then there exists  $\rho > 0$  such that  $\|u_n\|_\infty, \|u\|_\infty \leq \rho$  for every  $n$ .

By the Carathéodory property,

$$g(t, u_n(t)) \rightarrow g(t, u(t)), \quad \text{for a.e. } t \in (0, T),$$

and

$$\|g(t, u_n(t)) - g(t, u(t))\|_Y \leq 2\alpha_\rho(t), \quad \text{for a.e. } t \in (0, T).$$

Since  $\alpha_\rho \in L^1(0, T)$ , the dominated convergence theorem gives

$$\|N_g(u_n) - N_g(u)\|_{L^1([0, T]; Y)} \rightarrow 0.$$

Thus  $N_g$  is continuous.

Now let  $B \subset C([0, T]; X)$  be bounded. Then there exists  $\rho > 0$  such that  $\|u\|_\infty \leq \rho$  for every  $u \in B$ . Hence,

$$\|N_g(u)(t)\|_Y \leq \alpha_\rho(t)$$

for a.e.  $t \in (0, T)$  and every  $u \in B$ . Since  $\alpha_\rho \in L^1(0, T)$ , this implies that  $N_g(B)$  is equi-integrable.  $\square$

## 2.3. Bochner and variable-exponent spaces

Let  $X$  be a Banach space,  $I = [0, T]$ , and let  $1 \leq p \leq \infty$ . The *Bochner space*  $L^p(I; X)$  is the Banach-valued analogue of the usual Lebesgue space: its elements are strongly measurable functions (i.e., a.e. limits of  $X$ -valued simple functions, as required for Bochner integration) such that  $\|u(\cdot)\|_X \in L^p(I)$ . We also define

$$W^{1,p}(I; X) := \{u \in L^p(I; X) : u' \in L^p(I; X)\},$$

where the derivative is understood in the distributional sense.

**Proposition 2.3.1** *Let  $X$  be a Banach space and let  $1 \leq p \leq \infty$ . Then  $L^p(I; X)$  and  $W^{1,p}(I; X)$  are Banach spaces. Moreover,*

$$W^{1,p}(I; X) \hookrightarrow C([0, T]; X)$$

continuously. If  $X$  is reflexive and  $1 < p < \infty$ , then both spaces are reflexive.

For further details on Bochner spaces and Sobolev spaces with values in Banach spaces, see [2, 3].

We also recall the *variable-exponent spaces* used in the Galerkin approach. These spaces are part of the broader theory of nonstandard growth and are closely related to Orlicz-type settings. For background on Orlicz spaces and variable-exponent spaces, we refer to [5, 10, 20, 21].

Let  $p \in C([0, T])$  satisfy

$$1 < p_- := \min_{t \in [0, T]} p(t) \leq p(t) \leq p_+ := \max_{t \in [0, T]} p(t) < \infty,$$

and define  $p'(t) := p(t)/(p(t) - 1)$ . For a strongly measurable function  $u : I \rightarrow X$ , set

$$\rho_{p(\cdot)}(u) := \int_0^T \|u(t)\|_X^{p(t)} dt.$$

The *variable-exponent Bochner space*  $L^{p(\cdot)}(I; X)$  is endowed with the Luxemburg norm

$$\|u\|_{L^{p(\cdot)}(I; X)} := \inf \left\{ \lambda > 0 : \rho_{p(\cdot)}\left(\frac{u}{\lambda}\right) \leq 1 \right\}.$$

We define  $W^{1, p(\cdot)}(I; X)$  in the usual way by requiring  $u, u' \in L^{p(\cdot)}(I; X)$ .

**Proposition 2.3.2** *Let  $X$  be a reflexive Banach space. Then  $L^{p(\cdot)}(I; X)$  and  $W^{1, p(\cdot)}(I; X)$  are reflexive Banach spaces. Moreover,*

$$L^{p(\cdot)}(I; X) \hookrightarrow L^1(I; X), \quad W^{1, p(\cdot)}(I; X) \hookrightarrow C([0, T]; X)$$

continuously.

**Proposition 2.3.3** (Hölder inequality) *There exists  $C > 0$  such that, for every  $u \in L^{p(\cdot)}(I; X)$  and every  $v \in L^{p'(\cdot)}(I; X^*)$ ,*

$$\int_0^T \|u(t)\|_X \|v(t)\|_{X^*} dt \leq C \|u\|_{L^{p(\cdot)}(I; X)} \|v\|_{L^{p'(\cdot)}(I; X^*)}.$$

**Proposition 2.3.4** (Modular–norm relations) *Let  $u \in L^{p(\cdot)}(I; X)$ , then the following inequalities hold:*

$$\begin{cases} \|u\|_{L^{p(\cdot)}(I; X)}^{p_-} \leq \rho_{p(\cdot)}(u) \leq \|u\|_{L^{p(\cdot)}(I; X)}^{p_+}, & \text{if } \|u\|_{L^{p(\cdot)}(I; X)} > 1, \\ \|u\|_{L^{p(\cdot)}(I; X)}^{p_+} \leq \rho_{p(\cdot)}(u) \leq \|u\|_{L^{p(\cdot)}(I; X)}^{p_-}, & \text{if } \|u\|_{L^{p(\cdot)}(I; X)} \leq 1. \end{cases}$$

**Proposition 2.3.5** (Nemytskii operators in variable-exponent spaces) *Let  $X$  be a real separable reflexive Banach space and let  $p \in C([0, T])$  satisfy*

$$1 < p_- \leq p(t) \leq p_+ < \infty.$$

*Let  $A : [0, T] \times X \rightarrow X^*$  be a Carathéodory operator. Assume that there exist a constant  $C > 0$  and a function  $a \in L^{p'(\cdot)}([0, T])$  such that*

$$\|A(t, x)\|_{X^*} \leq a(t) + C\|x\|_X^{p(t)-1} \quad (2.1)$$

*for a.e.  $t \in [0, T]$  and every  $x \in X$ . Then the associated Nemytskii operator  $N_A : L^{p(\cdot)}([0, T]; X) \rightarrow L^{p'(\cdot)}([0, T]; X^*)$  defined by  $N_A(u)(t) := A(t, u(t))$  is well defined, maps bounded sets into bounded sets, and is continuous.*

PROOF. The measurability of  $N_A(u)$  follows from the Carathéodory property of  $A$  and the strong measurability of  $u$ . Moreover, by the growth assumption and the fact that  $(p(t) - 1)p'(t) = p(t)$ , there exists a constant  $C_1 > 0$  such that

$$\|A(t, u(t))\|_{X^*}^{p'(t)} \leq C_1 \left( a(t)^{p'(t)} + \|u(t)\|_X^{p(t)} \right) \quad (2.2)$$

for a.e.  $t \in [0, T]$ . Hence  $N_A(u) \in L^{p'(\cdot)}([0, T]; X^*)$ . The same estimate, together with the modular–norm relations, shows that  $N_A$  maps bounded subsets of  $L^{p(\cdot)}([0, T]; X)$  into bounded subsets of  $L^{p'(\cdot)}([0, T]; X^*)$ .

It remains to prove continuity. Let  $u_n \rightarrow u$  in  $L^{p(\cdot)}([0, T]; X)$ . By the usual subsequence criterion, it is enough to show that every subsequence has a further subsequence for which the convergence holds.

Thus, after passing to a subsequence, still denoted by  $(u_n)_n$ , we may assume that  $u_n(t) \rightarrow u(t)$  in  $X$  for a.e.  $t \in [0, T]$ , and that there exists  $g \in L^{p(\cdot)}([0, T])$  such that

$$\|u_n(t)\|_X + \|u(t)\|_X \leq g(t) \quad \text{for a.e. } t \in [0, T] \text{ and every } n. \quad (2.3)$$

By the Carathéodory property of  $A$ , we have

$$A(t, u_n(t)) \rightarrow A(t, u(t)) \quad \text{in } X^* \quad \text{for a.e. } t \in [0, T]. \quad (2.4)$$

Moreover, the growth assumption gives

$$\|A(t, u_n(t)) - A(t, u(t))\|_{X^*}^{p'(t)} \leq C \left( a(t)^{p'(t)} + g(t)^{p(t)} \right). \quad (2.5)$$



The last function belongs to  $L^1(0, T)$ . Hence, by the dominated convergence theorem,

$$\rho_{p'(\cdot)}(N_A(u_n) - N_A(u)) = \int_0^T \|A(t, u_n(t)) - A(t, u(t))\|_{X^*}^{p'(t)} dt \rightarrow 0. \quad (2.6)$$

By the modular–norm relations,

$$N_A(u_n) \rightarrow N_A(u) \quad \text{in } L^{p'(\cdot)}([0, T]; X^*). \quad (2.7)$$

Since every subsequence admits a further subsequence with this property, the whole sequence converges. Therefore  $N_A$  is continuous.  $\square$

## 2.4. Monotone operators

Let  $X$  be a real Banach space. An operator  $A : X \rightarrow X^*$  is called *monotone* if

$$\langle A(u) - A(v), u - v \rangle \geq 0 \quad \text{for every } u, v \in X,$$

and *strictly monotone* if the inequality is strict whenever  $u \neq v$ . It is called *coercive* if

$$\frac{\langle A(u), u \rangle}{\|u\|_X} \rightarrow \infty \quad \text{as } \|u\|_X \rightarrow \infty,$$

and *hemicontinuous* if, for every  $u, v, w \in X$ , the map  $s \mapsto \langle A(u + sv), w \rangle$  is continuous on  $\mathbb{R}$ . It is *demicontinuous* if  $x_n \rightarrow x$  in  $X$  implies  $A(x_n) \rightarrow A(x)$  in  $X^*$ .

**Theorem 2.4.1** (Minty–Browder) *Let  $X$  be a real reflexive Banach space and let  $A : X \rightarrow X^*$  be monotone, coercive, and hemicontinuous. Then  $A$  is surjective. If, in addition,  $A$  is strictly monotone, then  $A$  is bijective.*

For further details on monotone operators in Banach spaces and related evolution equations, see [2–4].

We will also use the following consequence of a Bolzano-type theorem for demicontinuous monotone operators. Since monotone hemicontinuous operators are demicontinuous by [4, Corollary 11.1], it follows from [18, Theorem 4].

**Lemma 2.4.2** *Let  $X$  be a reflexive Banach space and let  $F : X \rightarrow X^*$  be hemicontinuous. Assume that  $F$  is strictly monotone and that*

$$\langle F(x), x \rangle \rightarrow +\infty \quad \text{as } \|x\|_X \rightarrow \infty.$$

*Then there exists a unique  $x_0 \in X$  such that  $F(x_0) = 0$ .*

# Chapter 3

## The main problem and its abstract formulation

We consider the periodic problem

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T). \end{cases} \quad (3.1)$$

Here  $X$  is a real separable reflexive Banach space,  $X^*$  denotes its dual space,  $S : [0, T] \times X \rightarrow X^*$  is a nonlinear time-dependent operator, and  $f : [0, T] \times X \rightarrow X^*$  is a given nonlinear term. The operator  $S$  will be assumed to satisfy suitable monotonicity, coercivity, and continuity conditions, which will be specified below.

By a solution of (3.1), we mean a function  $u \in C^1([0, T]; X)$  such that  $S(\cdot, u'(\cdot)) \in W^{1,1}([0, T]; X^*)$  and the differential equation holds for a.e.  $t \in [0, T]$ .

The purpose of this chapter is to derive an equivalent formulation of (3.1). The main difficulty comes from the periodic boundary conditions. Indeed, after integrating the differential equation once, one obtains an unknown constant in  $X^*$ , which has to be chosen so that the condition  $u(0) = u(T)$  is satisfied. The remaining periodicity condition, namely  $u'(0) = u'(T)$ , is recovered through the compatibility condition obtained by integrating the equation over one period.

To handle these two constraints, we construct an abstract nonlinear Green operator associated with  $S$ . This operator allows us to separate the constant component of a periodic solution from its remaining part. More precisely, writing  $u = a + v$ , where  $a \in X$  and  $v \in C([0, T]; X)$  satisfies  $v(0) = v(T) = 0$ , the original periodic problem is transformed into a generalized Lyapunov–Schmidt system for the pair  $(a, v)$ .

This formulation provides the common framework for the mean-admissible and Galerkin approaches developed in the subsequent chapters.

### 3.1. Structural assumptions on the operator $S$

We assume that  $S : [0, T] \times X \rightarrow X^*$  satisfies the following conditions:

**(S0)** The mapping  $S : [0, T] \times X \rightarrow X^*$  is continuous,  $S(t, x) = 0$  if and only if  $x = 0$ , and  $S(0, x) = S(T, x)$  for every  $x \in X$ .

**(S1)** For every  $M > 0$ , there exists a continuous non-decreasing function  $\omega_M : [0, \infty) \rightarrow [0, \infty)$  such that

$$\omega_M(0) = 0, \quad \omega_M(r) > 0 \quad \text{for every } r > 0,$$

and

$$\langle S(t, x) - S(t, y), x - y \rangle \geq \omega_M(\|x - y\|_X)$$

for every  $t \in [0, T]$  and every  $x, y \in X$  satisfying  $\|x\|_X, \|y\|_X \leq M$ .

Moreover, there exists a continuous non-decreasing function  $\alpha : [0, \infty) \rightarrow [0, \infty)$  such that

$$\alpha(r) \xrightarrow{r \rightarrow \infty} \infty, \quad \langle S(t, x), x \rangle \geq \alpha(\|x\|_X) \|x\|_X \quad (3.2)$$

for every  $t \in [0, T]$  and every  $x \in X$ .

**(S2)** There exists a continuous function  $\beta : [0, T] \times [0, \infty) \rightarrow [0, \infty)$  such that

$$\|S(t, x)\|_{X^*} \leq \beta(t, \|x\|_X), \quad \text{for every } t \in [0, T], x \in X.$$

Under **(S0)**,  $t \mapsto S(t, u(t))$  is measurable for every measurable  $u : [0, T] \rightarrow X$ ; see [1, Lemma 4.51]. Moreover, **(S1)** implies that  $S(t, \cdot)$  is strictly monotone and coercive uniformly with respect to  $t$ , while **(S2)** ensures that  $S$  maps bounded sets into bounded sets uniformly with respect to  $t$ .

**Remark 3.1.1** Let  $X$  be a real Hilbert space, let  $\mathcal{R} : X \rightarrow X^*$  be the Riesz isomorphism, and define

$$S(t, 0) := 0, \quad S(t, x) := \|x\|_X^{p(t)-2} \mathcal{R}x, \quad x \neq 0,$$

where  $p \in C([0, T])$ ,  $p_- := \min_{t \in [0, T]} p(t) > 1$ ,  $p_+ := \max_{t \in [0, T]} p(t) < \infty$  and  $p(0) = p(T)$ . Then  $S$  satisfies **(S0)**–**(S2)**.

In the next proposition, we show that for each  $t \in [0, T]$  the mapping  $S(t, \cdot) : X \rightarrow X^*$  has an inverse. In this case, we denote by  $\Gamma : [0, T] \times X^* \rightarrow X$  the map  $\Gamma(t, x) = (S(t, \cdot))^{-1}(x)$ .

**Proposition 3.1.2** *Let  $S : [0, T] \times X \rightarrow X^*$  satisfy **(S0)** – **(S2)**, then for every  $t \in [0, T]$  the mapping  $S(t, \cdot) : X \rightarrow X^*$  is bijective, and the related mapping  $\Gamma(t, \cdot) : X^* \rightarrow X$  is strictly monotone. Furthermore, the mapping  $\Gamma : [0, T] \times X^* \rightarrow X$  is continuous.*

PROOF. For each fixed  $t \in [0, T]$ , the map  $S(t, \cdot) : X \rightarrow X^*$  is continuous, and therefore hemi-continuous. Moreover, by (S1) and (3.2), it is strictly monotone and coercive. Hence, by the Minty–Browder Theorem (see Theorem 2.4.1),  $S(t, \cdot) : X \rightarrow X^*$  is surjective. Since  $S(t, \cdot)$  is also injective by (S1), it follows that  $S(t, \cdot)$  is bijective, and hence  $\Gamma$  is well defined.

The monotonicity of  $\Gamma(t, \cdot)$  for every  $t \in [0, T]$  follows directly from (S1), since for every  $x, y \in X^*$  with  $x \neq y$  we have

$$\langle x - y, \Gamma(t, x) - \Gamma(t, y) \rangle = \langle S(t, \Gamma(t, x)) - S(t, \Gamma(t, y)), \Gamma(t, x) - \Gamma(t, y) \rangle > 0.$$

It remains to prove the joint continuity of  $\Gamma$ . Let  $(t_n, y_n) \rightarrow (t, y)$  in  $[0, T] \times X^*$ , and set  $x_n := \Gamma(t_n, y_n)$ ,  $x := \Gamma(t, y)$ . From (3.2),

$$\alpha(\|x_n\|_X)\|x_n\|_X \leq \langle y_n, x_n \rangle \leq \|y_n\|_{X^*}\|x_n\|_X,$$

so  $(x_n)_n$  is bounded in  $X$ . Choose  $M > 0$  such that  $\|x_n\|_X, \|x\|_X \leq M$  for every  $n$ . Hence, using (S1),

$$\omega_M(\|x_n - x\|_X) \leq \langle y_n - y, x_n - x \rangle + \langle S(t, x) - S(t_n, x), x_n - x \rangle.$$

The right-hand side tends to zero, since  $(x_n)_n$  is bounded,  $y_n \rightarrow y$ , and  $S(\cdot, x)$  is continuous. Thus  $\omega_M(\|x_n - x\|_X) \rightarrow 0$ , and therefore  $x_n \rightarrow x$  in  $X$  (since  $\omega_M$  is non-decreasing and  $\omega_M(r) > 0$  for every  $r > 0$ ). This proves the joint continuity of  $\Gamma$ .

□

**Proposition 3.1.3** *Under (S0) – (S2),  $\Gamma : [0, T] \times X^* \rightarrow X$  satisfies the following properties:*

- (i)  $\|\Gamma(t, y)\|_X \rightarrow \infty$  as  $\|y\|_{X^*} \rightarrow \infty$  uniformly in  $t \in [0, T]$ .
- (ii) If  $u \in C([0, T]; X)$  (resp.  $z \in C([0, T]; X^*)$ ), then the map  $t \mapsto S(t, u(t))$  belongs to  $C([0, T]; X^*)$ , and in particular  $S(\cdot, u(\cdot)) \in L^p([0, T]; X^*)$ ,  $\forall 1 \leq p \leq \infty$ . (similarly the map  $t \mapsto \Gamma(t, z(t))$  belongs to  $C([0, T]; X)$ , and in particular  $\Gamma(\cdot, z(\cdot)) \in L^p([0, T]; X)$ ,  $\forall 1 \leq p \leq \infty$ ).
- (iii) If  $(g_n)_n \subset C([0, T]; X^*)$  is a convergent sequence, say  $g_n \rightarrow g$  as  $n \rightarrow \infty$ , then  $\Gamma(t, g_n(t)) \rightarrow \Gamma(t, g(t))$  in  $C([0, T]; X)$ .

PROOF. For (i), suppose that there exist  $(t_n, y_n)_n \subset [0, T] \times X^*$  such that  $\|y_n\|_{X^*} \rightarrow \infty$  and  $\|\Gamma(t_n, y_n)\|_X \leq M$ . If  $x_n := \Gamma(t_n, y_n)$ , then  $y_n = S(t_n, x_n)$ . Since  $(t_n, x_n)_n$  is bounded, (S2) implies that  $(y_n)_n$  is bounded, a contradiction.

Statement (ii) follows from the joint continuity of  $S$  and  $\Gamma$ . Since  $[0, T]$  is compact, the corresponding functions are bounded, and therefore belong to the corresponding  $L^p$ -spaces for every  $1 \leq p \leq \infty$ .

To prove (iii), set

$$m_n := \sup_{t \in [0, T]} \|\Gamma(t, g_n(t)) - \Gamma(t, g(t))\|_X.$$

If  $m_n \not\rightarrow 0$ , then there exist  $\varepsilon > 0$ , a subsequence  $(g_{n_k})_k$ , and points  $t_k \in [0, T]$  such that

$$\|\Gamma(t_k, g_{n_k}(t_k)) - \Gamma(t_k, g(t_k))\|_X \geq \varepsilon.$$

Passing to a further subsequence, we may assume that  $t_k \rightarrow t_0$ . Since  $g_{n_k} \rightarrow g$  uniformly and  $g$  is continuous, we have

$$g_{n_k}(t_k) \rightarrow g(t_0), \quad g(t_k) \rightarrow g(t_0).$$

The joint continuity of  $\Gamma$  then gives

$$\Gamma(t_k, g_{n_k}(t_k)) \rightarrow \Gamma(t_0, g(t_0)), \quad \Gamma(t_k, g(t_k)) \rightarrow \Gamma(t_0, g(t_0)),$$

contradicting the previous inequality. Hence  $m_n \rightarrow 0$ , and (iii) follows.  $\square$

## 3.2. Solving an implicit equation

The existence analysis for the periodic problem follows from this observation. Let  $u : [0, T] \rightarrow X$  be a solution of

$$\begin{cases} \frac{d}{dt} S(t, u'(t)) = h(t), \\ u(0) = u(T), \end{cases} \quad (3.3)$$

where  $h \in L^1([0, T]; X^*)$ . At this stage, we only impose the condition  $u(0) = u(T)$ . This condition determines the integration constant in  $X^*$ . The second periodicity condition, namely  $u'(0) = u'(T)$ , will be recovered later through another condition.

Integrating over  $[0, t]$ , with  $t \in [0, T]$ , we obtain

$$S(t, u'(t)) = c + \int_0^t h(s) ds$$

where  $c = S(0, u'(0))$ . Applying  $\Gamma(t, \cdot)$  to this identity yields

$$u'(t) = \Gamma(t, c + H(h)(t)),$$

where  $H(h)(t) = \int_0^t h(s) ds$ . Integrating one more time over  $[0, t]$ , we arrive at

$$u(t) = u(0) + \int_0^t \Gamma(s, c + H(h)(s)) ds.$$

Since  $u(T) = u(0)$ , it necessarily follows that

$$\int_0^T \Gamma(t, c + H(h)(t)) dt = 0. \quad (3.4)$$

Thus, the element  $c \in X^*$  is precisely one that enforces the condition  $u(0) = u(T)$ . Conversely, if  $u$  is defined by

$$u(t) = a + \int_0^t \Gamma(s, c + H(h)(s)) ds,$$

for some  $a \in X$  and some  $c \in X^*$  satisfying (3.4), then  $u$  is a solution of (3.3). Therefore, the problem reduces to proving that there exists a unique element  $c \in X^*$  satisfying (3.4).

Since  $H(h) \in C([0, T]; X^*)$ , it is convenient to study the following more general implicit equation. Given  $g \in C([0, T]; X^*)$ , define

$$\Phi_g : X^* \rightarrow X, \quad \Phi_g(c) := \int_0^T \Gamma(t, c + g(t)) dt.$$

With this notation, condition (3.4) becomes

$$\Phi_{H(h)}(c) = 0.$$

Thus, the construction of the solution of (3.3) reduces to proving the existence and uniqueness of a zero of  $\Phi_g$ .

We can now establish the solvability of the implicit equation  $\Phi_g(c) = 0$ .

**Lemma 3.2.1** *Under (S0) – (S2), if  $g \in C([0, T]; X^*)$ , then the map  $\Phi_g$  is continuous and there exists a unique  $c^* \in X^*$  such that  $\Phi_g(c^*) = 0$  (in particular  $\Phi_0(0) = 0$ ).*

*Furthermore, if we define  $\widehat{c} : C([0, T]; X^*) \rightarrow X^*$  such that  $\Phi_g(\widehat{c}(g)) = 0$ , then  $\widehat{c}$  sends bounded sets into bounded sets and is continuous.*

**PROOF.** *Existence and uniqueness.* The continuity of  $\Phi_g : X^* \rightarrow X$  follows by Proposition 3.1.3, (iii). Define

$$\Psi := J \circ \Phi_g : X^* \rightarrow X^{**},$$

where  $J : X \rightarrow X^{**}$  is the canonical embedding. Since  $X$  is reflexive,  $X^*$  is reflexive as well. Moreover, since  $\Phi_g$  is continuous,  $\Psi$  is continuous, and in particular hemicontinuous.

By the definition of the embedding, we have

$$\langle \Psi(c_1) - \Psi(c_2), c_1 - c_2 \rangle = \langle J(\Phi_g(c_1) - \Phi_g(c_2)), c_1 - c_2 \rangle = \langle c_1 - c_2, \Phi_g(c_1) - \Phi_g(c_2) \rangle > 0,$$

since by the strict monotonicity of  $\Gamma(t, \cdot)$ ,

$$\langle c_1 - c_2, \Phi_g(c_1) - \Phi_g(c_2) \rangle = \int_0^T \langle c_1 - c_2, \Gamma(t, c_1 + g(t)) - \Gamma(t, c_2 + g(t)) \rangle dt > 0.$$

Thus the map  $\Psi$  is strictly monotone. We want to apply Lemma 2.4.2 to conclude the existence of a solution  $c \in X^*$  of  $\Phi_g(c) = 0$ , for this we just have to prove that

$$\langle \Psi(c), c \rangle \rightarrow \infty \quad \text{as} \quad \|c\|_{X^*} \rightarrow \infty. \quad (3.5)$$

Notice that, by (3.2), for every  $y \in X^*$  and every  $t \in [0, T]$ , we have

$$\langle y, \Gamma(t, y) \rangle \geq \alpha (\|\Gamma(t, y)\|_X) \|\Gamma(t, y)\|_X. \quad (3.6)$$

Hence,

$$\begin{aligned} \langle \Psi(c), c \rangle &= \langle c, \Phi_g(c) \rangle = \int_0^T \langle c, \Gamma(t, c + g(t)) \rangle dt \\ &= \int_0^T \langle c + g(t), \Gamma(t, c + g(t)) \rangle dt - \int_0^T \langle g(t), \Gamma(t, c + g(t)) \rangle dt \\ &\geq \int_0^T [\alpha (\|\Gamma(t, c + g(t))\|_X) - \|g\|_\infty] \|\Gamma(t, c + g(t))\|_X dt, \end{aligned}$$

Now, since  $\|c\|_{X^*} \rightarrow \infty$  implies that  $\|c + g(t)\|_{X^*} \rightarrow \infty$  uniformly for  $t \in [0, T]$ , Proposition 3.1.3 yields  $\|\Gamma(t, c + g(t))\|_X \rightarrow \infty$  uniformly for  $t \in [0, T]$ . Consequently,  $\alpha (\|\Gamma(t, c + g(t))\|_X) \rightarrow \infty$  as  $\|c\|_{X^*} \rightarrow \infty$ , uniformly in  $t \in [0, T]$ . Therefore,

$$\int_0^T [\alpha (\|\Gamma(t, c + g(t))\|_X) - \|g\|_\infty] \|\Gamma(t, c + g(t))\|_X dt \rightarrow \infty, \quad \text{as } \|c\|_{X^*} \rightarrow \infty.$$

This proves that  $\langle \Psi(c), c \rangle \rightarrow \infty$  as  $\|c\|_{X^*} \rightarrow \infty$ , and hence (3.5) follows.

By applying Lemma 2.4.2 to the map  $\Psi$ , it follows that there exists a unique  $c^* \in X^*$  such that  $\Psi(c^*) = 0$  and hence that  $\Phi_g(c^*) = 0$ .

Notice that by **(S0)** it follows that  $\Gamma(t, c) = 0$  if and only if  $c = 0$ , and hence it follows that:

$$\Phi_0(0) = \int_0^T \Gamma(t, 0) dt = 0,$$

and therefore  $\Phi_0(c) = 0$  if and only if  $c = 0$ .

*Boundedness on bounded sets.* Recall now that the map  $\widehat{c} : C([0, T]; X^*) \rightarrow X^*$  is defined by  $\Phi_g(\widehat{c}(g)) = 0$ . To prove that  $\widehat{c}$  sends bounded sets into bounded sets, let  $\Lambda \subset C([0, T]; X^*)$  be a

bounded set, say  $\|l\|_\infty \leq M$  for all  $l \in \Lambda$ , for some  $M > 0$ . Then by the definition of  $\widehat{c}(l)$  we have:

$$\int_0^T \langle \widehat{c}(l), \Gamma(t, \widehat{c}(l) + l(t)) \rangle dt = 0,$$

and thus:

$$\int_0^T \langle \widehat{c}(l) + l(t), \Gamma(t, \widehat{c}(l) + l(t)) \rangle dt = \int_0^T \langle l(t), \Gamma(t, \widehat{c}(l) + l(t)) \rangle dt.$$

If we suppose by contradiction that  $\{\widehat{c}(l) : l \in \Lambda\}$  is not bounded, then we can find some  $l \in \Lambda$  with  $\|\widehat{c}(l)\|_{X^*}$  sufficiently large such that, by (i) of Proposition 3.1.3,  $\|\Gamma(t, \widehat{c}(l) + l(t))\|_X$  is sufficiently large such that

$$\alpha(\|\Gamma(t, \widehat{c}(l) + l(t))\|_X) \geq 2M, \quad \text{uniformly in } t \in [0, T].$$

Hence,

$$\begin{aligned} 2M \int_0^T \|\Gamma(t, \widehat{c}(l) + l(t))\|_X dt &\leq \int_0^T \alpha(\|\Gamma(t, \widehat{c}(l) + l(t))\|_X) \|\Gamma(t, \widehat{c}(l) + l(t))\|_X dt \\ &\leq \int_0^T \langle \widehat{c}(l) + l(t), \Gamma(t, \widehat{c}(l) + l(t)) \rangle dt = \int_0^T \langle l(t), \Gamma(t, \widehat{c}(l) + l(t)) \rangle dt, \\ &\leq M \int_0^T \|\Gamma(t, \widehat{c}(l) + l(t))\|_X dt \end{aligned}$$

Since  $\|\widehat{c}(l)\|_{X^*}$  is chosen sufficiently large, the last integral is strictly positive. Dividing by it, we obtain  $2M \leq M$ , which is impossible because  $M > 0$ . This contradiction proves that  $\widehat{c}(\Lambda)$  is bounded in  $X^*$ . Therefore,  $\widehat{c}$  sends bounded sets into bounded sets.

*Continuity of  $\widehat{c}$ .* Let  $g_n \rightarrow g$  in  $C([0, T]; X^*)$ , and set  $c_n := \widehat{c}(g_n)$  and  $c := \widehat{c}(g)$ . Since  $\widehat{c}$  maps bounded sets into bounded sets, the sequence  $(c_n)_n$  is bounded in  $X^*$ . Fix  $R > 0$  such that  $c_n, c \in B_R(0)$  for all  $n$ , and choose  $M_g > 0$  such that  $\|g_n\|_\infty, \|g\|_\infty \leq M_g$  for all  $n$ .

For  $s \in [0, T]$ , define

$$\Delta_n(s) := \Gamma(s, c_n + g_n(s)) - \Gamma(s, c + g(s)).$$

Since  $c_n, c \in B_R(0)$  and  $\|g_n\|_\infty, \|g\|_\infty \leq M_g$ , we have

$$\|c_n + g_n(s)\|_{X^*}, \|c + g(s)\|_{X^*} \leq R + M_g$$

for every  $s \in [0, T]$  and every  $n$ . It follows by (3.6) that

$$\alpha(\|\Gamma(s, c_n + g_n(s))\|_X), \alpha(\|\Gamma(s, c + g(s))\|_X) \leq R + M_g.$$



Since  $\alpha(r) \rightarrow \infty$  as  $r \rightarrow \infty$ , there exists  $M_B > 0$ , depending only on  $R$  and  $M_g$ , such that

$$\|\Gamma(s, c_n + g_n(s))\|_X, \|\Gamma(s, c + g(s))\|_X \leq M_B, \quad \forall s \in [0, T], n \in \mathbb{N}.$$

Therefore,

$$\|\Delta_n(s)\|_X \leq 2M_B, \quad \forall s \in [0, T], n \in \mathbb{N}.$$

We claim that

$$\int_0^T \omega_{M_B}(\|\Delta_n(s)\|_X) ds \xrightarrow{n \rightarrow \infty} 0. \quad (3.7)$$

Indeed, since  $\Phi_{g_n}(c_n) = 0$  and  $\Phi_g(c) = 0$ , taking duality pairings with  $c_n - c$  and subtracting gives the identity

$$\int_0^T \langle (c_n + g_n(s)) - (c + g(s)), \Delta_n(s) \rangle ds = \int_0^T \langle g_n(s) - g(s), \Delta_n(s) \rangle ds. \quad (3.8)$$

By **(S1)**, for each  $s \in [0, T]$ ,

$$\langle (c_n + g_n(s)) - (c + g(s)), \Delta_n(s) \rangle \geq \omega_{M_B}(\|\Delta_n(s)\|_X).$$

Using the estimate

$$\langle g_n(s) - g(s), \Delta_n(s) \rangle \leq \|g_n(s) - g(s)\|_{X^*} \|\Delta_n(s)\|_X \leq \|g_n - g\|_\infty \|\Delta_n(s)\|_X,$$

and combining it with (3.8), we obtain

$$\int_0^T \omega_{M_B}(\|\Delta_n(s)\|_X) ds \leq \int_0^T \|g_n(s) - g(s)\|_{X^*} \|\Delta_n(s)\|_X ds \leq 2M_B T \|g_n - g\|_\infty \xrightarrow{n \rightarrow \infty} 0,$$

which proves (3.7). Consequently,

$$\omega_{M_B}(\|\Delta_n(\cdot)\|_X) \rightarrow 0 \quad \text{in } L^1([0, T]; \mathbb{R}),$$

and the same holds for every subsequence  $(n_k)_k$ . By [3, Theorem 4.9], there exists a sub-subsequence  $(n_{k_j})_j$  such that

$$\omega_{M_B}(\|\Delta_{n_{k_j}}(s)\|_X) \rightarrow 0 \quad \text{for a.e. } s \in [0, T].$$

In particular, we may choose  $t_0 \in [0, T]$  such that

$$\omega_{M_B}(\|\Delta_{n_{k_j}}(t_0)\|_X) \rightarrow 0.$$

As in the proof of Proposition 3.1.2, the properties of  $\omega_{M_B}$  imply that  $\|\Delta_{n_{k_j}}(t_0)\|_X \rightarrow 0$  as  $j \rightarrow \infty$ , and hence

$$\Gamma(t_0, c_{n_{k_j}} + g_{n_{k_j}}(t_0)) \rightarrow \Gamma(t_0, c + g(t_0)) \quad \text{in } X.$$

Using the continuity of  $S(t_0, \cdot)$ , we deduce

$$c_{n_{k_j}} + g_{n_{k_j}}(t_0) \rightarrow c + g(t_0) \quad \text{in } X^*.$$

Since  $g_{n_{k_j}} \rightarrow g$  in  $C([0, T]; X^*)$ , in particular  $g_{n_{k_j}}(t_0) \rightarrow g(t_0)$  in  $X^*$ , and therefore  $c_{n_{k_j}} \rightarrow c$  in  $X^*$ .

We have shown that every subsequence of  $(c_n)_n$  admits a further subsequence converging to  $c$ . Hence  $c_n \rightarrow c$  in  $X^*$ , i.e.,

$$\widehat{c}(g_n) \rightarrow \widehat{c}(g),$$

which proves the continuity of  $\widehat{c}$ .

□

### 3.3. The generalized Lyapunov–Schmidt system

To study the problem (3.1) we will consider the *non-homogeneous periodic boundary value problem*:

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = h(t), & t \in (0, T), \\ u(0) = u(T), & u'(0) = u'(T), \end{cases} \quad (3.9)$$

where  $h \in L^1([0, T]; X^*)$ . Our goal is to derive representation formulas for its solutions and, on this basis, define the corresponding abstract nonlinear Green operator.

Before stating the representation result, we introduce the operators that will be used in the abstract formulation. Define

$$H : L^1([0, T]; X^*) \rightarrow C([0, T]; X^*), \quad H(h)(t) := \int_0^t h(s) ds,$$

and

$$Q : L^1([0, T]; X^*) \rightarrow X^*, \quad Qh := \int_0^T h(t) dt.$$

We use the notation

$$\begin{aligned} C_0([0, T]; X) &:= \{v \in C([0, T]; X) : v(0) = v(T) = 0\}, \\ C_T^1([0, T]; X) &:= \{u \in C^1([0, T]; X) : u(0) = u(T), u'(0) = u'(T)\}. \end{aligned}$$

We also define

$$\iota : C([0, T]; X) \rightarrow C([0, T]; X), \quad \iota(u)(t) := u(0).$$

Finally, using the map  $\widehat{c}$  introduced in Lemma 3.2.1, define

$$\mathcal{T} : L^1([0, T]; X^*) \rightarrow C_0([0, T]; X)$$

by

$$\mathcal{T}(h)(t) := \int_0^t \Gamma(s, \widehat{c}(H(h)) + H(h)(s)) \, ds.$$

Notice that  $\mathcal{T}(h) \in C_0([0, T]; X)$  since  $\mathcal{T}(h)(0) = 0$  and, by the definition of  $\widehat{c}$ ,

$$\mathcal{T}(h)(T) = \int_0^T \Gamma(s, \widehat{c}(H(h)) + H(h)(s)) \, ds = 0.$$

Moreover, since the integrand is continuous, we also have  $\mathcal{T}(h) \in C^1([0, T]; X)$ .

**Lemma 3.3.1** *Let  $S$  satisfy (S0)–(S2), and let  $h \in L^1([0, T]; X^*)$ . Then the non-homogeneous periodic problem (3.9) has a solution  $u \in C_T^1([0, T]; X)$  if and only if*

$$Qh := \int_0^T h(t) \, dt = 0.$$

*In this case, the solution is unique up to an additive constant and is given by*

$$u(t) = u(0) + \int_0^t \Gamma(s, \widehat{c}(H(h)) + H(h)(s)) \, ds. \quad (3.10)$$

*Equivalently, the non-homogeneous periodic problem (3.9) admits a solution  $u \in C_T^1([0, T]; X)$  if and only if  $u$  satisfies*

$$\begin{cases} u = \iota(u) + \mathcal{T}(h), \\ Qh = 0. \end{cases} \quad (3.11)$$

**PROOF.** Assume first that  $u \in C_T^1([0, T]; X)$  is a solution of (3.9). Integrating the differential equation over  $[0, T]$  and using (S0) and  $u'(0) = u'(T)$ , we obtain

$$Qh = \int_0^T h(t) \, dt = S(T, u'(T)) - S(0, u'(0)) = S(0, u'(T)) - S(0, u'(0)) = 0.$$

Integrating now the differential equation over  $[0, t]$ , we obtain

$$S(t, u'(t)) = c + H(h)(t),$$

where  $c := S(0, u'(0))$ . Applying  $\Gamma(t, \cdot)$  yields

$$u'(t) = \Gamma(t, c + H(h)(t)).$$

Since  $u(0) = u(T)$ , integrating this identity over  $[0, T]$  gives

$$0 = u(T) - u(0) = \int_0^T \Gamma(t, c + H(h)(t)) dt.$$

Therefore,

$$\Phi_{H(h)}(c) = 0.$$

By Lemma 3.2.1, the solution of this equation is unique. Consequently,

$$c = \widehat{c}(H(h)).$$

Integrating the expression for  $u'$  over  $[0, t]$ , we obtain

$$u(t) = u(0) + \int_0^t \Gamma(s, \widehat{c}(H(h)) + H(h)(s)) ds.$$

This proves the representation formula (3.10).

Conversely, suppose that  $Qh = 0$ . Let  $a \in X$  and define

$$u(t) := a + \int_0^t \Gamma(s, \widehat{c}(H(h)) + H(h)(s)) ds.$$

By the definition of  $\widehat{c}$ , we have

$$\int_0^T \Gamma(s, \widehat{c}(H(h)) + H(h)(s)) ds = 0.$$

Hence  $u(T) = a = u(0)$ . Moreover, differentiating, applying  $S(t, \cdot)$ , and then differentiating again, we obtain

$$\frac{d}{dt} S(t, u'(t)) = h(t)$$

for a.e.  $t \in [0, T]$ .

It remains to verify the periodicity of the derivative. Since  $Qh = 0$ , we have

$$H(h)(T) = \int_0^T h(t) dt = 0 = H(h)(0).$$

Furthermore, **(S0)** implies that

$$S(0, x) = S(T, x) \quad \text{for every } x \in X.$$

Since  $S(0, \cdot)$  and  $S(T, \cdot)$  are bijective, it follows that

$$\Gamma(0, y) = \Gamma(T, y) \quad \text{for every } y \in X^*.$$

Consequently, since  $H(h)(T) = H(h)(0) = 0$ ,

$$u'(T) = \Gamma(T, \widehat{c}(H(h)) + H(h)(T)) = \Gamma(0, \widehat{c}(H(h)) + H(h)(0)) = u'(0).$$

Thus,  $u$  is a solution of (3.9).

Finally, let  $u_1$  and  $u_2$  be two solutions of (3.9). By the previous argument, both solutions satisfy

$$u'_i(t) = \Gamma(t, \widehat{c}(H(h)) + H(h)(t)), \quad i = 1, 2.$$

Hence  $u'_1(t) = u'_2(t)$  for every  $t \in [0, T]$ , and therefore  $u_1 - u_2$  is constant. This proves uniqueness up to an additive constant.  $\square$

We now apply the previous representation result to the nonlinear periodic problem

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad (3.12)$$

where  $f : [0, T] \times X \rightarrow X^*$  is a Carathéodory operator whose Nemytskii operator is  $N_f : C([0, T]; X) \rightarrow L^1([0, T]; X^*)$ . By Lemma 3.3.1, using  $h = N_f(u)$ , a function  $u$  is a solution of (3.12) if and only if

$$\begin{cases} u = \iota(u) + \mathcal{T}(N_f(u)), \\ QN_f(u) = 0. \end{cases} \quad (3.13)$$

By setting  $a := u(0)$  and  $v := \mathcal{T}(N_f(u))$ , then  $v \in C_0([0, T]; X)$  and  $u = a + v$ . Therefore (3.13) is equivalent to the system

$$\begin{cases} v = \mathcal{T}(N_f(a + v)), \\ QN_f(a + v) = 0. \end{cases} \quad (3.14)$$

Thus, solving the original periodic problem is equivalent to finding  $a \in X$  and  $v \in C_0([0, T]; X)$  satisfying (3.14). The generalized Lyapunov–Schmidt system suggests two natural strategies. The first consists in solving the averaged equation with respect to the constant component  $a$ . Under the mean-admissibility condition, this reduces the problem to a single fixed point equation.

The terminology *generalized Lyapunov–Schmidt system* reflects the analogy between the decomposition  $u = a + v$  and the classical Lyapunov–Schmidt splitting. Unlike the classical method, however, the reduced equation  $QN_f(a + v) = 0$  remains infinite-dimensional whenever  $X$  is infinite-dimensional.

The second strategy treats both equations simultaneously through Galerkin approximations and topological degree, for nonlinearities of the form  $f = m + k$ .

# Chapter 4

## Existence for *mean-admissible* nonlinearities

The generalized Lyapunov–Schmidt system obtained in the previous chapter is given by

$$\begin{cases} v = \mathcal{T}(N_f(a + v)), \\ QN_f(a + v) = 0. \end{cases}$$

Unlike in the classical Lyapunov–Schmidt reduction, the constraint

$$QN_f(a + v) = 0$$

is not necessarily finite-dimensional. Therefore, it is natural to look for an additional mechanism allowing us to solve this equation with respect to  $a$  in terms of the component  $v$ . In this way, the generalized Lyapunov–Schmidt system can be reduced to a single fixed point problem.

The main result of this chapter is a continuation theorem for this reduced problem. More precisely, we introduce a class of nonlinearities, called *mean-admissible*, for which the averaged equation can be solved uniquely with respect to the constant component  $a$ . In the presence of a compact Gelfand triple

$$X \hookrightarrow H \hookrightarrow X^*,$$

the nonlinear Green operator becomes compact when viewed with values in  $C_0([0, T]; H)$ . This allows us to apply the Leray–Schauder degree to the reduced equation and obtain an  $H$ -valued periodic solution of the original problem. The final part of the chapter gives an application in which the nonlinearity is *mean-admissible* but not pointwise monotone nor pointwise coercive.

### 4.1. The class of *mean-admissible* operators

**Definition 4.1.1** Let  $Y, Z$  be real separable reflexive Banach spaces and let  $f : [0, T] \times Y \rightarrow Z$  be a Carathéodory operator such that its Nemytskii operator maps  $C([0, T]; Y)$  into  $L^1([0, T]; Z)$ . For

every  $v \in C_0([0, T]; Y)$ , define

$$F_v : Y \rightarrow Z, \quad F_v(a) := \int_0^T f(t, a + v(t)) dt.$$

We say that  $f$  belongs to the *mean-admissible class* if there exists a continuous map

$$\widehat{A}_f : C_0([0, T]; Y) \rightarrow Y$$

such that:

1.  $\widehat{A}_f(v)$  is the unique solution of:

$$F_v(a) = 0.$$

2.  $\widehat{A}_f$  maps bounded sets into bounded sets.

This class is motivated by the construction of the nonlinear Green operator for the periodic problem presented in Lemma 3.2.1. In that case, the periodic condition leads to the implicit equation

$$\int_0^T \Gamma(t, \widehat{c}(g) + g(t)) dt = 0.$$

This produces a map  $\widehat{c}$  naturally associated with the inverse  $\Gamma(t, \cdot)$  of  $S(t, \cdot)$ . The *mean-admissibility* condition abstracts this mechanism for the nonlinearity  $f$ : it ensures that the constraint  $QN_f(a+v) = 0$  can be solved with respect to  $a$ , reducing the periodic problem to a single fixed point equation.

The following result provides a practical criterion for verifying that a nonlinearity  $f : [0, T] \times Y \rightarrow Y^*$  belongs to the *mean-admissible* class.

**Lemma 4.1.2** *Let  $Y$  be a real separable reflexive Banach space and let  $f : [0, T] \times Y \rightarrow Y^*$  be a continuous operator. Assume that:*

1. *There exists a non-decreasing function  $\eta : [0, \infty) \rightarrow [0, \infty)$  such that*

$$\eta(r) > 0 \quad \text{for every } r > 0,$$

*and*

$$\langle f(t, x) - f(t, y), x - y \rangle \geq \eta(\|x - y\|_Y)$$

*for every  $t \in [0, T]$  and every  $x, y \in Y$ .*

2. *There exist functions  $\Theta : [0, \infty) \rightarrow \mathbb{R}$  and  $\gamma : [0, \infty) \rightarrow [0, \infty)$ , with  $\gamma$  non-decreasing, such that*

$$\Theta(r) \rightarrow \infty \quad \text{as } r \rightarrow \infty,$$

$$\langle f(t, x), x \rangle \geq \Theta(\|x\|_Y) \|x\|_Y, \quad \|f(t, x)\|_{Y^*} \leq \gamma(\|x\|_Y),$$

for every  $t \in [0, T]$  and every  $x \in Y$ , and

$$\Theta(r)r - M\gamma(r) \rightarrow \infty \quad \text{as } r \rightarrow \infty$$

for every  $M > 0$ .

Then  $f$  belongs to the mean-admissible class.

PROOF. For  $g \in C([0, T]; Y)$ , define

$$\Phi_g(c) := \int_0^T f(t, c + g(t)) dt.$$

*Continuity, existence and uniqueness.* Let  $(c_n)_n \subset Y$  be such that  $c_n \rightarrow c$  in  $Y$ . Since  $(c_n)_n$  is bounded, there exists  $R > 0$  such that

$$\|c_n + g(t)\|_Y, \|c + g(t)\|_Y \leq R$$

for every  $t \in [0, T]$  and every  $n$ . Hence,

$$\|f(t, c_n + g(t))\|_{Y^*}, \|f(t, c + g(t))\|_{Y^*} \leq \gamma(R).$$

Since  $f(t, c_n + g(t)) \rightarrow f(t, c + g(t))$  in  $Y^*$  for every  $t \in [0, T]$ , the dominated convergence theorem yields

$$\Phi_g(c_n) \rightarrow \Phi_g(c) \quad \text{in } Y^*.$$

Thus,  $\Phi_g$  is continuous. In particular,  $\Phi_g$  is hemicontinuous.

Moreover, for every  $c_1, c_2 \in Y$  with  $c_1 \neq c_2$ , we have

$$\begin{aligned} \langle \Phi_g(c_1) - \Phi_g(c_2), c_1 - c_2 \rangle &= \int_0^T \langle f(t, c_1 + g(t)) - f(t, c_2 + g(t)), c_1 - c_2 \rangle dt \\ &\geq T\eta(\|c_1 - c_2\|_Y) > 0. \end{aligned}$$

Therefore,  $\Phi_g$  is strictly monotone.

We now prove that

$$\langle \Phi_g(c), c \rangle \rightarrow \infty \quad \text{as } \|c\|_Y \rightarrow \infty. \quad (4.1)$$

Indeed,

$$\begin{aligned} \langle \Phi_g(c), c \rangle &= \int_0^T \langle f(t, c + g(t)), c + g(t) \rangle dt - \int_0^T \langle f(t, c + g(t)), g(t) \rangle dt \\ &\geq \int_0^T [\Theta(\|c + g(t)\|_Y)\|c + g(t)\|_Y - \|g\|_\infty \gamma(\|c + g(t)\|_Y)] dt. \end{aligned}$$



Since  $\|c + g(t)\|_Y \rightarrow \infty$  uniformly in  $t \in [0, T]$  as  $\|c\|_Y \rightarrow \infty$ , the last integral tends to infinity. Hence, (4.1) follows.

By Lemma 2.4.2, there exists a unique  $c \in Y$  such that  $\Phi_g(c) = 0$ . We define

$$\widehat{A}_f(g) := c.$$

*Boundedness on bounded sets.* Let  $\Lambda \subset C([0, T]; Y)$  be bounded, and choose  $M > 0$  such that

$$\|g\|_\infty \leq M \quad \text{for every } g \in \Lambda.$$

Suppose, by contradiction, that  $\{\widehat{A}_f(g) : g \in \Lambda\}$  is not bounded. Then there exists a sequence  $(g_n)_n \subset \Lambda$  such that

$$\|\widehat{A}_f(g_n)\|_Y \rightarrow \infty.$$

Since

$$\|\widehat{A}_f(g_n) + g_n(t)\|_Y \geq \|\widehat{A}_f(g_n)\|_Y - M,$$

we have

$$\|\widehat{A}_f(g_n) + g_n(t)\|_Y \rightarrow \infty$$

uniformly in  $t \in [0, T]$ . On the other hand, since  $\Phi_{g_n}(\widehat{A}_f(g_n)) = 0$ , we obtain

$$\begin{aligned} 0 &= \int_0^T \langle f(t, \widehat{A}_f(g_n) + g_n(t)), \widehat{A}_f(g_n) \rangle dt \\ &= \int_0^T \langle f(t, \widehat{A}_f(g_n) + g_n(t)), \widehat{A}_f(g_n) + g_n(t) \rangle dt \\ &\quad - \int_0^T \langle f(t, \widehat{A}_f(g_n) + g_n(t)), g_n(t) \rangle dt \\ &\geq \int_0^T \left[ \Theta(\|\widehat{A}_f(g_n) + g_n(t)\|_Y) \|\widehat{A}_f(g_n) + g_n(t)\|_Y - M\gamma(\|\widehat{A}_f(g_n) + g_n(t)\|_Y) \right] dt. \end{aligned}$$

The last integral tends to infinity, which is a contradiction. Therefore,  $\widehat{A}_f$  maps bounded sets into bounded sets.

*Continuity of  $\widehat{A}_f$ .* Let  $(g_n)_n \subset C([0, T]; Y)$  be such that  $g_n \rightarrow g$  in  $C([0, T]; Y)$ , and set

$$a_n := \widehat{A}_f(g_n), \quad a := \widehat{A}_f(g).$$

Since  $\widehat{A}_f$  maps bounded sets into bounded sets, the sequence  $(a_n)_n$  is bounded in  $Y$ . Hence, there exists  $R > 0$  such that

$$\|a_n\|_Y, \|a\|_Y, \|g_n\|_\infty, \|g\|_\infty \leq R$$

for every  $n$ . Adding and subtracting  $\Phi_{g_n}(a)$ , and using  $\Phi_{g_n}(a_n) = \Phi_g(a) = 0$ , we obtain

$$\langle \Phi_{g_n}(a_n) - \Phi_{g_n}(a), a_n - a \rangle = -\langle \Phi_{g_n}(a) - \Phi_g(a), a_n - a \rangle.$$

Therefore,

$$T\eta(\|a_n - a\|_Y) \leq \langle \Phi_{g_n}(a_n) - \Phi_{g_n}(a), a_n - a \rangle \leq \|\Phi_{g_n}(a) - \Phi_g(a)\|_{Y^*} \|a_n - a\|_Y.$$

Since  $(a_n - a)_n$  is bounded, there exists  $C > 0$  such that

$$T\eta(\|a_n - a\|_Y) \leq C\|\Phi_{g_n}(a) - \Phi_g(a)\|_{Y^*}.$$

Furthermore,

$$f(t, a + g_n(t)) \rightarrow f(t, a + g(t)) \quad \text{in } Y^*$$

for every  $t \in [0, T]$ , and

$$\|f(t, a + g_n(t))\|_{Y^*}, \|f(t, a + g(t))\|_{Y^*} \leq \gamma(2R).$$

By the dominated convergence theorem,

$$\Phi_{g_n}(a) \rightarrow \Phi_g(a) \quad \text{in } Y^*.$$

Consequently,

$$\eta(\|a_n - a\|_Y) \rightarrow 0.$$

Since  $\eta$  is non-decreasing and  $\eta(r) > 0$  for every  $r > 0$ , it follows that

$$\|a_n - a\|_Y \rightarrow 0.$$

Thus,  $\widehat{A}_f$  is continuous. □

We now present a simple example illustrating the previous criterion.

**Example 4.1.3** Let  $\Omega \subset \mathbb{R}^N$  be a bounded domain, let  $Y = L^q(\Omega)$ , and let  $2 < p \leq q < \infty$ . Let  $m \in C([0, T])$  satisfy

$$0 < m_0 \leq m(t) \leq m_1,$$

and let  $b \in C([0, T]; L^{q'}(\Omega))$ , where  $q' = q/(q - 1)$ . Define

$$f : [0, T] \times L^q(\Omega) \rightarrow L^{q'}(\Omega)$$

by

$$f(t, u) := m(t)|u|^{q-2}u + |u|^{p-2}u + b(t).$$

Then  $f$  satisfies the hypotheses of Lemma 4.1.2. Indeed, since the map  $s \mapsto |s|^{q-2}s$  is uniformly monotone, there exists  $C_q > 0$  such that

$$\begin{aligned}\langle f(t, u) - f(t, v), u - v \rangle &\geq m_0 \int_{\Omega} (|u|^{q-2}u - |v|^{q-2}v)(u - v) dx \\ &\geq m_0 C_q \|u - v\|_{L^q}^q.\end{aligned}$$

Thus, condition (1) holds with

$$\eta(r) := m_0 C_q r^q.$$

Set  $B := \|b\|_{\infty}$ . Then

$$\begin{aligned}\langle f(t, u), u \rangle &= m(t) \|u\|_{L^q}^q + \|u\|_{L^p}^p + \langle b(t), u \rangle \\ &\geq (m_0 \|u\|_{L^q}^{q-1} - B) \|u\|_{L^q}.\end{aligned}$$

Hence, we may take

$$\Theta(r) := m_0 r^{q-1} - B.$$

Moreover, since  $p \leq q$  and  $\Omega$  is bounded,

$$\||u|^{p-2}u\|_{L^{q'}} \leq C \|u\|_{L^q}^{p-1}.$$

Therefore,

$$\|f(t, u)\|_{L^{q'}} \leq m_1 \|u\|_{L^q}^{q-1} + C \|u\|_{L^q}^{p-1} + B.$$

Thus, we may take

$$\gamma(r) := m_1 r^{q-1} + C r^{p-1} + B.$$

Finally, for every  $M > 0$ ,

$$\Theta(r)r - M\gamma(r) = m_0 r^q - Br - M(m_1 r^{q-1} + C r^{p-1} + B) \rightarrow \infty$$

as  $r \rightarrow \infty$ . Therefore, all the hypotheses of Lemma 4.1.2 are satisfied.

The class of nonlinearities described in Lemma 4.1.2 is contained in the *mean-admissible* class. The following example shows that this inclusion is strict.

**Example 4.1.4** (A *mean-admissible* operator which is not pointwise monotone nor coercive) Let  $T = 2\pi$ , let  $H$  be a real Hilbert space,  $b \in C([0, 2\pi]; H)$ , and

$$m(t) := 1 + 2 \sin t, \quad f(t, x) := m(t)x + b(t).$$

Since  $\int_0^{2\pi} m(t) dt = 2\pi$ , for each  $v \in C_0([0, 2\pi]; H)$  the equation

$$F_v(a) := \int_0^{2\pi} f(t, a + v(t)) dt = 0$$

has the unique solution

$$\widehat{A}_f(v) = -\frac{1}{2\pi} \left( \int_0^{2\pi} m(t)v(t) dt + \int_0^{2\pi} b(t) dt \right).$$

It follows directly from this formula that  $\widehat{A}_f$  is continuous and maps bounded sets into bounded sets. Therefore  $f$  belongs to the *mean-admissible* class. However, since  $m(t) < 0$  for some  $t$ ,

$$(f(t, x) - f(t, y), x - y)_H = m(t)\|x - y\|_H^2 < 0$$

for  $x \neq y$ , and also

$$(f(t, x), x)_H = m(t)\|x\|_H^2 + (b(t), x)_H \rightarrow -\infty$$

as  $\|x\|_H \rightarrow \infty$ . Hence  $f(t, \cdot)$  is not pointwise monotone nor pointwise coercive.

## 4.2. Recovering compactness through a Gelfand triple

The generalized Lyapunov–Schmidt system derived in the previous chapter is valid in the abstract Banach-space setting. However, in order to apply the Leray–Schauder degree, we need compactness of the associated fixed point operator. Since this compactness is not automatic in infinite-dimensional Banach spaces, we now impose an additional structural assumption.

We assume that there exists a real Hilbert space  $H$  such that

$$X \hookrightarrow H \hookrightarrow X^*$$

is a Gelfand triple, with  $X \hookrightarrow H$  compact and dense. We identify  $H$  with its dual and regard  $H$  as continuously embedded into  $X^*$ .

In this chapter the nonlinearity is defined on  $H$  and takes values in  $H$ . Consequently, the solutions obtained below are naturally understood as  $H$ -valued periodic solutions. More precisely, if  $u = a + v$ , then the constant component  $a$  belongs to  $H$ , while the oscillatory component is generated by the nonlinear Green operator. Thus  $v$  has the regularity provided by  $\mathcal{T}$ ; in particular,

$$v \in C_0([0, T]; X), \quad v' \in C([0, T]; X).$$

Thus, throughout this chapter, a solution of the periodic problem means a function  $u \in$

$C_T([0, T]; H)$  of the form

$$u = a + v, \quad a \in H, \quad v \in C_0([0, T]; X) \cap C^1([0, T]; X),$$

such that

$$\frac{d}{dt}S(t, v'(t)) = i^* f(t, u(t)) \quad \text{in } X^*$$

for a.e.  $t \in [0, T]$ , together with the periodic condition

$$v'(0) = v'(T).$$

Here  $i^* : H \hookrightarrow X^*$  denotes the canonical embedding.

The equivalence in Lemma 3.3.1 remains valid for functions  $u = a + v$  with  $a \in H$  and  $v \in C_0([0, T]; X) \cap C^1([0, T]; X)$ , since  $u' = v'$  and the proof applies verbatim with  $h = i^* N_f(u)$ .

Since  $\mathcal{T}(h) \in C_0([0, T]; X)$  for every  $h \in L^1([0, T]; X^*)$ , the compact embedding  $X \hookrightarrow H$  allows us to regard the nonlinear Green operator as

$$\mathcal{T} : L^1([0, T]; X^*) \rightarrow C_0([0, T]; H).$$

The role of the Gelfand triple is precisely to recover compactness when  $\mathcal{T}$  is viewed with values in  $C_0([0, T]; H)$ .

The following lemma is the key compactness result for  $\mathcal{T}$ .

**Lemma 4.2.1** *Assume that there exists a real Hilbert space  $H$  such that  $X \hookrightarrow H \hookrightarrow X^*$  is a Gelfand triple, with  $X \hookrightarrow H$  compact and dense. Then the operator*

$$\mathcal{T} : L^1([0, T]; X^*) \rightarrow C_0([0, T]; H)$$

*is continuous and maps bounded sets into relatively compact sets.*

**PROOF.** The continuity of  $\mathcal{T}$  follows from the continuity of the operators involved in its definition. Indeed, if  $h_n \rightarrow h$  in  $L^1([0, T]; X^*)$ , then

$$H(h_n) \rightarrow H(h) \quad \text{in } C([0, T]; X^*).$$

By the continuity of  $\widehat{c}$  and Proposition 3.1.3, we obtain

$$\Gamma(\cdot, \widehat{c}(H(h_n)) + H(h_n)(\cdot)) \rightarrow \Gamma(\cdot, \widehat{c}(H(h)) + H(h)(\cdot)) \quad \text{in } C([0, T]; X).$$

Integrating over  $[0, t]$ , uniformly in  $t \in [0, T]$ , gives

$$\mathcal{T}(h_n) \rightarrow \mathcal{T}(h) \quad \text{in } C_0([0, T]; H).$$

Let  $\Lambda \subset L^1([0, T]; X^*)$  be bounded and let  $(h_n)_n \subset \Lambda$ . Then there exists  $B > 0$  such that

$$\|h_n\|_{L^1([0, T]; X^*)} \leq B \quad \text{for every } n.$$

Therefore,  $(H(h_n))_n$  is bounded in  $C([0, T]; X^*)$ . Since  $\widehat{c}$  maps bounded sets into bounded sets, the sequence

$$z_n(t) := \widehat{c}(H(h_n)) + H(h_n)(t)$$

is bounded in  $C([0, T]; X^*)$ . Hence, there exists  $R > 0$  such that

$$\|z_n(t)\|_{X^*} \leq R \quad \text{for every } t \in [0, T] \text{ and every } n.$$

By (3.6), we have

$$\alpha(\|\Gamma(t, z_n(t))\|_X) \leq R.$$

Since  $\alpha(r) \rightarrow \infty$  as  $r \rightarrow \infty$ , there exists  $C > 0$  such that

$$\|\Gamma(t, z_n(t))\|_X \leq C \quad \text{for every } t \in [0, T] \text{ and every } n.$$

Consequently, for  $t, t' \in [0, T]$ ,

$$\|\mathcal{T}(h_n)(t) - \mathcal{T}(h_n)(t')\|_H \leq C_{X,H} \int_t^{t'} \|\Gamma(s, z_n(s))\|_X ds \leq C_{X,H} C |t' - t|.$$

Thus,  $(\mathcal{T}(h_n))_n$  is equicontinuous in  $C([0, T]; H)$ . Moreover, for every fixed  $t \in [0, T]$ ,

$$\|\mathcal{T}(h_n)(t)\|_X \leq \int_0^t \|\Gamma(s, z_n(s))\|_X ds \leq CT.$$

Hence,  $\{\mathcal{T}(h_n)(t) : n \in \mathbb{N}\}$  is bounded in  $X$  and therefore relatively compact in  $H$ , because  $X \hookrightarrow H$  is compact.

By the generalized Arzelà–Ascoli theorem,  $(\mathcal{T}(h_n))_n$  admits a subsequence converging in  $C([0, T]; H)$ . Therefore,  $\mathcal{T}(\Lambda)$  is relatively compact in  $C_0([0, T]; H)$ .  $\square$

We now introduce the convention that will be used throughout the rest of this chapter. A map  $f : [0, T] \times H \rightarrow H$  will be called an *admissible Carathéodory* operator if it is measurable with respect to  $t$  for every  $x \in H$ , continuous with respect to  $x$  for a.e.  $t \in [0, T]$ , and satisfies the following local growth condition: for every  $\rho > 0$  there exists  $\eta_\rho \in L^1(0, T)$  such that

$$\|f(t, x)\|_H \leq \eta_\rho(t)$$

for a.e.  $t \in (0, T)$  and every  $x \in H$  with  $\|x\|_H \leq \rho$ .

By Proposition 2.2.2, the associated Nemytskii operator

$$N_f : C([0, T]; H) \rightarrow L^1([0, T]; H), \quad N_f(u)(t) := f(t, u(t)),$$

is continuous and maps bounded sets into equi-integrable, and in particular bounded, subsets of  $L^1([0, T]; H)$ . Through the embedding  $H \hookrightarrow X^*$ , we may regard  $N_f$  as an  $X^*$ -valued operator.

Accordingly, the generalized Lyapunov–Schmidt system takes the form

$$\begin{cases} v = \mathcal{T}(N_f(a + v)), \\ QN_f(a + v) = 0, \end{cases} \quad a \in H, \quad v \in C_0([0, T]; H). \quad (4.2)$$

Although the system is written in  $C_0([0, T]; H)$ , any solution of the first equation has the additional regularity provided by the nonlinear Green operator. In particular, the component  $v$  belongs to  $C_0([0, T]; X)$ .

If  $f$  belongs to the *mean-admissible* class, the second equation in (4.2) is solved by

$$a = \widehat{A}_f(v).$$

Thus, the system is reduced to the fixed point equation

$$v = \mathcal{T} \circ N_f \circ (\widehat{A}_f + I)(v), \quad v \in C_0([0, T]; H). \quad (4.3)$$

In summary, the *mean-admissibility* condition allows us to solve the second equation in (4.2), reducing the periodic problem to the fixed point equation (4.3). On the other hand, Lemma 4.2.1 provides the compactness required to apply the Leray–Schauder degree.

### 4.3. A continuation theorem

We denote by

$$C_T([0, T]; H) := \{u \in C([0, T]; H) : u(0) = u(T)\}.$$

With this, and the last two results shown, we can derive a continuation theorem.

**Theorem 4.3.1** *Let  $S$  satisfy (S0)–(S2), and assume that there exists a real Hilbert space  $H$  such that  $X \hookrightarrow H \hookrightarrow X^*$  is a Gelfand triple, with  $X \hookrightarrow H$  compact and dense. Let  $\Omega \subset C_T([0, T]; H)$  be a bounded open set, and let  $f : [0, T] \times H \rightarrow H$  be an admissible Carathéodory operator. Assume furthermore that:*

1.  *$f$  belongs to the mean-admissible class;*

2. there exists  $c \in H$  such that the constant function  $t \mapsto c$  belongs to  $\Omega$  and

$$\int_0^T f(t, c) dt = 0;$$

3. for every  $\lambda \in (0, 1)$ , the problem

$$\begin{cases} \frac{d}{dt} S(t, u'(t)) = \lambda i^* f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T), \end{cases} \quad (4.4)$$

has no  $H$ -valued periodic solution on  $\partial\Omega$ .

Then the problem

$$\begin{cases} \frac{d}{dt} S(t, u'(t)) = i^* f(t, u(t)), \\ u(0) = u(T), \quad u'(0) = u'(T). \end{cases} \quad (4.5)$$

admits an  $H$ -valued periodic solution  $u \in \overline{\Omega}$ .

PROOF. We first observe that, if (4.4) has a solution  $u \in \partial\Omega$  and  $\lambda = 1$ , then the conclusion of the theorem follows immediately. Therefore, we may assume from now on that (4.4) has no solution on  $\partial\Omega$  for  $\lambda = 1$ .

Define

$$M := \left\{ u \in C_T([0, T]; H) : \int_0^T f(t, u(t)) dt = 0 \right\},$$

which is a closed subset of  $C_T([0, T]; H)$ , and define

$$R(v) := \widehat{A}_f(v) + v, \quad \Omega_0 := R^{-1}(\Omega \cap M) \subset C_0([0, T]; H).$$

The map  $R$  is a homeomorphism from  $C_0([0, T]; H)$  onto  $M$ . Indeed, by the definition of  $\widehat{A}_f$ , we have  $R(v) \in M$  for every  $v \in C_0([0, T]; H)$ . Conversely, if  $u = R(v)$ , then, since  $v(0) = 0$ ,

$$u(0) = \widehat{A}_f(v), \quad v = u - u(0).$$

Hence  $R^{-1}(u) = u - u(0)$ , which is continuous.

Since  $\Omega \cap \mathcal{M}$  is relatively open in  $\mathcal{M}$  and  $R : C_0([0, T]; H) \rightarrow \mathcal{M}$  is a homeomorphism,  $\Omega_0 = R^{-1}(\Omega \cap \mathcal{M})$  is open. Moreover,  $\Omega_0$  is bounded because, for every  $v \in \Omega_0$ ,

$$v = R(v) - R(v)(0), \quad \|v\|_\infty \leq 2\|R(v)\|_\infty,$$

and  $R(v) \in \Omega$ . By hypothesis, there exists  $c \in H$  such that the constant function  $t \mapsto c$  belongs to



$\Omega$  and

$$\int_0^T f(t, c) dt = 0.$$

By uniqueness in the definition of  $\widehat{A}_f$ , we have  $\widehat{A}_f(0) = c$ . Therefore,  $0 \in \Omega_0$ .

For  $\lambda \in [0, 1]$ , define

$$\mathcal{G}(\lambda, v) := \mathcal{T}(\lambda N_f(R(v))).$$

By Proposition 2.2.2,  $N_f$  is continuous and maps bounded sets into bounded subsets of  $L^1([0, T]; H)$ . Since  $R$  is continuous and maps bounded sets into bounded sets, and since  $\mathcal{T}$  is continuous and maps bounded subsets of  $L^1([0, T]; X^*)$  into relatively compact subsets of  $C_0([0, T]; H)$  by Lemma 4.2.1, it follows that  $\mathcal{G}$  is completely continuous.

We claim that

$$v \neq \mathcal{G}(\lambda, v) \quad \text{for every } v \in \partial\Omega_0 \text{ and every } \lambda \in [0, 1].$$

Let first  $\lambda \in (0, 1]$ , and suppose, by contradiction, that there exists  $v \in \partial\Omega_0$  such that

$$v = \mathcal{G}(\lambda, v).$$

Set  $u := R(v)$ . Since  $R$  takes values in  $M$ , we have

$$QN_f(u) = \int_0^T f(t, u(t)) dt = 0.$$

Moreover,

$$v = \mathcal{T}(\lambda N_f(u)).$$

Therefore, by the abstract formulation,  $u$  solves (4.4).

It remains to prove that  $u \in \partial\Omega$ . Since  $\Omega_0 = R^{-1}(\Omega \cap M)$  and  $\Omega_0$  is open, the condition  $v \in \partial\Omega_0$  implies that  $v \notin \Omega_0$ . Hence,

$$u = R(v) \notin \Omega \cap M.$$

Since  $u \in M$ , we conclude that  $u \notin \Omega$ . On the other hand, there exists a sequence  $(v_n)_n \subset \Omega_0$  such that  $v_n \rightarrow v$ . By the continuity of  $R$ ,

$$R(v_n) \rightarrow R(v) = u.$$

Since  $R(v_n) \in \Omega \cap M \subset \Omega$ , it follows that

$$u \in \overline{\Omega} \setminus \Omega = \partial\Omega.$$

For  $\lambda \in (0, 1)$ , this contradicts the a priori assumption. For  $\lambda = 1$ , it contradicts the initial reduction, namely that (4.5) has no solution on  $\partial\Omega$ .

For  $\lambda = 0$ , we have

$$\mathcal{G}(0, v) = \mathcal{T}(0) = 0$$

for every  $v \in C_0([0, T]; H)$ . Hence, the only fixed point is  $v = 0$ . Since  $0 \in \Omega_0$  and  $\Omega_0$  is open, we have

$$0 \notin \partial\Omega_0.$$

Therefore,  $\mathcal{G}(0, \cdot)$  has no fixed point on  $\partial\Omega_0$ .

The Leray–Schauder degree is thus well defined and, by homotopy invariance,

$$d_{LS}[I - \mathcal{G}(1, \cdot), \Omega_0, 0] = d_{LS}[I - \mathcal{G}(0, \cdot), \Omega_0, 0] = d_{LS}[I, \Omega_0, 0] = 1.$$

Therefore, there exists  $v \in \Omega_0$  such that

$$v = \mathcal{G}(1, v).$$

Consequently,

$$u := R(v) = \widehat{A}_f(v) + v$$

is a solution of (4.5). Since  $v \in \Omega_0$ , we have

$$u \in \Omega \cap M \subset \Omega \subset \overline{\Omega}.$$

This concludes the proof. □

## 4.4. Application to a periodic problem with a sign-changing linear term

Let  $T = 2\pi$ , let  $\Omega \subset \mathbb{R}^N$  be a bounded domain with Lipschitz boundary, and let  $\beta > 2$ . Set

$$X := W_0^{1,\beta}(\Omega), \quad H := L^2(\Omega), \quad X^* := W^{-1,\beta'}(\Omega),$$

where  $\beta' = \beta/(\beta - 1)$ . Since  $\beta > 2 > \frac{2N}{N+2}$ , the embeddings  $X \hookrightarrow H \hookrightarrow X^*$  form a compact Gelfand triple.

Let

$$m(t) := 1 + 2 \sin t, \quad b \in C([0, 2\pi]; H),$$

and define

$$f(t, u) := m(t)u + b(t).$$

We consider the periodic problem

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = i^*(m(t)u(t) + b(t)) & \text{in } X^*, \\ u(0) = u(2\pi), \quad u'(0) = u'(2\pi), \end{cases} \quad (4.6)$$

where  $i^* : H \rightarrow X^*$  is the canonical embedding.

**Proposition 4.4.1** *Let  $2 < \mu \leq \nu$ , and assume that  $S : [0, 2\pi] \times X \rightarrow X^*$  satisfies (S0)–(S2). Suppose, in addition, that there exists  $C > 0$  such that*

$$\langle S(t, x), x \rangle \geq C \min \{ \|x\|_X^\mu, \|x\|_X^\nu \} \quad (4.7)$$

*for every  $t \in [0, 2\pi]$  and every  $x \in X$ . Then problem (4.6) admits at least one  $H$ -valued periodic solution.*

PROOF. By Example 4.1.4, the map  $f(t, u) = m(t)u + b(t)$  belongs to the *mean-admissible* class. We only need an a priori bound for the solutions of

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = \lambda i^* f(t, u(t)), \\ u(0) = u(2\pi), \quad u'(0) = u'(2\pi), \end{cases} \quad \lambda \in (0, 1]. \quad (4.8)$$

Let  $u$  be such a solution and set

$$v(t) := u(t) - u(0).$$

Then  $v(0) = v(2\pi) = 0$  and  $v' = u'$ . Testing the equation against  $v$ , integrating by parts, and using (4.7), we obtain

$$\int_0^{2\pi} \min \{ \|u'(t)\|_X^\mu, \|u'(t)\|_X^\nu \} dt \leq C \left( \|u\|_{L^2([0, 2\pi]; H)} + 1 \right) \|v\|_{L^2([0, 2\pi]; H)}. \quad (4.9)$$

Integrating the equation over one period gives

$$\int_0^{2\pi} f(t, u(t)) dt = 0.$$

Since  $u = u(0) + v$ , the formula from Example 4.1.4 yields

$$u(0) = \widehat{A}_f(v), \quad \|u(0)\|_H \leq C \|v\|_{L^2([0, 2\pi]; H)} + C.$$

By Poincaré's inequality and the embedding  $X \hookrightarrow H$ ,

$$\|v\|_{L^2([0,2\pi];H)} \leq C\|u'\|_{L^2([0,2\pi];X)}.$$

Consequently,

$$\|u\|_{L^2([0,2\pi];H)} \leq C\|u'\|_{L^2([0,2\pi];X)} + C.$$

Using this estimate in (4.9), we obtain

$$\int_0^{2\pi} \min \{ \|u'(t)\|_X^\mu, \|u'(t)\|_X^\nu \} dt \leq C \left( 1 + \|u'\|_{L^2([0,2\pi];X)}^2 \right).$$

Since  $2 < \mu \leq \nu$ ,

$$r^\mu \leq 1 + \min\{r^\mu, r^\nu\}, \quad r \geq 0,$$

and hence

$$\|u'\|_{L^\mu([0,2\pi];X)}^\mu \leq C \left( 1 + \|u'\|_{L^\mu([0,2\pi];X)}^2 \right).$$

Because  $\mu > 2$ , it follows that

$$\|u'\|_{L^\mu([0,2\pi];X)} + \|u'\|_{L^2([0,2\pi];X)} \leq C.$$

Therefore,

$$\|v\|_{C([0,2\pi];H)} + \|u\|_{C([0,2\pi];H)} \leq C.$$

Choosing a sufficiently large ball

$$\mathcal{O} := \{u \in C_T([0, 2\pi]; H) : \|u\|_{C([0,2\pi];H)} < \rho\},$$

the homotopy problems have no solutions on  $\partial\mathcal{O}$ , and the constant function  $t \mapsto \widehat{A}_f(0)$  belongs to  $\mathcal{O}$ . Theorem 4.3.1 therefore gives a solution of (4.6).  $\square$

The hypotheses of Proposition 4.4.1 hold for the following classes of operators.

**Example 4.4.2** (A double-phase-type operator) Let  $p \in C([0, 2\pi])$  satisfy

$$1 < p_- \leq p(t) \leq p_+ \leq \beta, \quad p(0) = p(2\pi).$$

For  $v, \phi \in X$ , define

$$\langle S(t, v), \phi \rangle := \int_{\Omega} \left( |\nabla v|^{p(t)-2} \nabla v + |\nabla v|^{\beta-2} \nabla v \right) \cdot \nabla \phi dx.$$

The  $p(t)$ -phase is monotone, while the  $\beta$ -phase satisfies

$$\langle S(t, v) - S(t, w), v - w \rangle \geq C_\beta \|v - w\|_X^\beta.$$

Thus the monotonicity part of **(S1)** holds with

$$\omega_M(r) := C_\beta r^\beta.$$

Moreover,

$$\langle S(t, v), v \rangle = \int_\Omega \left( |\nabla v|^{p(t)} + |\nabla v|^\beta \right) dx \geq \|v\|_X^\beta,$$

and

$$\|S(t, v)\|_{X^*} \leq C \left( 1 + \|v\|_X^{\beta-1} \right).$$

Hence (4.7) holds with

$$\mu = \nu = \beta,$$

and Proposition 4.4.1 applies.

**Example 4.4.3** (A variable-exponent duality-map operator) Consider

$$S(t, x) := \|x\|_X^{p(t)-2} j(x),$$

where  $j : X \rightarrow X^*$  is the normalized duality map and

$$\beta \leq p_- \leq p(t) \leq p_+ < \infty, \quad p(0) = p(2\pi).$$

If  $A_\beta : X \rightarrow X^*$  denotes the  $\beta$ -Laplacian operator, then

$$j(x) = \|x\|_X^{2-\beta} A_\beta(x), \quad S(t, x) = \|x\|_X^{p(t)-\beta} A_\beta(x).$$

Since  $X = W_0^{1,\beta}(\Omega)$ , endowed with the gradient norm, is isometric to a closed subspace of  $L^\beta(\Omega; \mathbb{R}^d)$ , it is uniformly convex and uniformly smooth. Hence since  $p(t) \geq \beta$ , the standard power-type monotonicity estimate for this type of spaces gives

$$\langle S(t, x) - S(t, y), x - y \rangle \geq C \min \left\{ \|x - y\|_X^{p_-}, \|x - y\|_X^{p_+} \right\}.$$

Thus the monotonicity part of **(S1)** holds. Furthermore,

$$\langle S(t, x), x \rangle = \|x\|_X^{p(t)} \geq \min \left\{ \|x\|_X^{p_-}, \|x\|_X^{p_+} \right\}, \quad \|S(t, x)\|_{X^*} = \|x\|_X^{p(t)-1} \leq 1 + \|x\|_X^{p_+-1}.$$

Therefore, (4.7) holds with  $\mu = p_-$ ,  $\nu = p_+$ , and Proposition 4.4.1 applies.

# Chapter 5

## A Hartman-type existence result

In the previous chapter, the *mean-admissibility* condition allowed us to solve the averaged equation and reduce the generalized Lyapunov–Schmidt system to a single fixed point problem.

The purpose of this chapter is to establish a Hartman-type existence result for nonlinearities of the form

$$f(t, x) = m(t, x) + k(t, x),$$

where  $m(t, \cdot)$  is monotone and  $k(t, \cdot)$  is sequentially weak-to-strong continuous. The full map  $f(t, \cdot)$  need not be monotone.

The underlying strategy is to approximate the infinite-dimensional problem by finite-dimensional ones, solve the projected equations, and then pass to the limit. Galerkin approximations and topological degree provide the approximate solutions, while a Minty argument, together with the structure of  $m + k$ , allows us to identify their weak limit as a solution of the original problem.

This formulation contains the purely monotone case by taking  $m = f$  and  $k \equiv 0$ . In that situation, monotonicity propagates the Hartman inequality from the boundary of a ball to its exterior, and additional uniqueness properties are obtained. On the other hand, when the full map is purely weak-to-strong, the exterior Hartman condition becomes degenerate in infinite-dimensional spaces: every periodic solution is stationary and the zero function is always a solution. This explains why the compactness assumption is imposed only on the perturbation  $k$ .

Within these structural assumptions, the results obtained below extend the finite-dimensional Hartman-type principles of Fan and Fan [7] and García-Huidobro, Manásevich, Mawhin, and Tanaka [8] to differential equations in reflexive Banach spaces.

After proving the general theorem and discussing its two limiting cases, we consider radial operators in Hilbert spaces and show that the solution obtained by the Galerkin construction remains inside the ball appearing in the Hartman condition. Finally, this result is applied to a spatially nonlocal Sobolev-type problem for which the corresponding pointwise bound is explicit.

## 5.1. Additional assumptions and functional setting

Throughout this chapter,  $X$  denotes a real separable reflexive Banach space. Let  $p \in C([0, T])$  satisfy

$$1 < p_- := \min_{t \in [0, T]} p(t) \leq p(t) \leq p_+ := \max_{t \in [0, T]} p(t) < \infty,$$

and let

$$p'(t) := \frac{p(t)}{p(t) - 1}$$

be its conjugate exponent.

We assume that  $S : [0, T] \times X \rightarrow X^*$  satisfies the following conditions for every  $t \in [0, T]$  and every  $x, y \in X$ :

**(H0)** The map  $S$  is continuous,  $S(t, x) = 0$  if and only if  $x = 0$ , and  $S(0, x) = S(T, x)$ .

**(H1)** For every  $M > 0$ , there exists a continuous non-decreasing function  $\omega_M : [0, \infty) \rightarrow [0, \infty)$  such that

$$\omega_M(0) = 0, \quad \omega_M(r) > 0 \quad \text{for every } r > 0,$$

and

$$\langle S(t, x) - S(t, y), x - y \rangle \geq \omega_M(\|x - y\|_X)$$

whenever  $\|x\|_X, \|y\|_X \leq M$ .

**(H2)** There exists  $c_0 > 0$  such that

$$\langle S(t, x), x \rangle \geq c_0 \|x\|_X^{p(t)}.$$

**(H3)** There exists  $C_S > 0$  such that

$$\|S(t, x)\|_{X^*} \leq C_S \left(1 + \|x\|_X^{p(t)-1}\right).$$

These assumptions are stronger than those imposed in Chapter 3. Indeed, **(H0)** coincides with **(S0)**, while **(H1)** gives the local uniform monotonicity required in **(S1)**. Moreover, **(H2)** implies

$$\langle S(t, x), x \rangle \geq c_0 \min \{ \|x\|_X^{p_-}, \|x\|_X^{p_+} \}.$$

Thus the coercivity condition of Chapter 3 holds with

$$\alpha(r) := c_0 \min \{ r^{p_- - 1}, r^{p_+ - 1} \}, \quad r \geq 0.$$

Finally, **(H3)** implies the growth condition **(S2)**. Hence, by the results of Chapter 3, the inverse

$$\Gamma(t, \cdot) := (S(t, \cdot))^{-1} : X^* \rightarrow X$$

is well defined and continuous.

Let  $m, k : [0, T] \times X \rightarrow X^*$  be Carathéodory maps and define

$$f(t, x) := m(t, x) + k(t, x).$$

We impose the local growth condition

**(F1)** For every  $r > 0$ , there exists  $\eta_r \in L^{p'(\cdot)}([0, T])$  such that

$$\|m(t, x)\|_{X^*} + \|k(t, x)\|_{X^*} \leq \eta_r(t)$$

for a.e.  $t \in [0, T]$  and every  $x \in X$  satisfying  $\|x\|_X \leq r$ .

We consider the periodic problem

$$\begin{cases} \frac{d}{dt}S(t, u'(t)) = m(t, u(t)) + k(t, u(t)), & t \in (0, T), \\ u(0) = u(T), & u'(0) = u'(T). \end{cases} \quad (5.1)$$

Throughout this chapter, by a *solution* of (5.1) we mean a function  $u \in C^1([0, T]; X)$  such that

$$S(\cdot, u'(\cdot)) \in W^{1, p'(\cdot)}([0, T]; X^*),$$

the differential equation holds for a.e.  $t \in [0, T]$ , and

$$u(0) = u(T), \quad u'(0) = u'(T).$$

We shall also use the corresponding weak formulation. A function  $u \in W^{1, p(\cdot)}([0, T]; X)$  is called a *weak solution* if

$$u(0) = u(T)$$

and

$$\int_0^T \langle S(t, u'(t)), \varphi'(t) \rangle dt + \int_0^T \langle m(t, u(t)) + k(t, u(t)), \varphi(t) \rangle dt = 0$$

for every  $\varphi \in W^{1, p(\cdot)}([0, T]; X)$  satisfying  $\varphi(0) = \varphi(T)$ .

Every solution is a weak solution, as follows directly from integration by parts. Conversely, under the assumptions of this chapter, every weak periodic solution obtained below has the regularity and satisfies the boundary conditions required in the preceding definition.

Set

$$V := \{u \in W^{1, p(\cdot)}([0, T]; X) : u(0) = u(T)\},$$



endowed with the norm

$$\|u\|_V := \|u\|_{L^{p(\cdot)}([0,T];X)} + \|u'\|_{L^{p(\cdot)}([0,T];X)}.$$

The space  $V$  is reflexive and the one-dimensional Sobolev embedding gives

$$V \hookrightarrow C([0, T]; X)$$

continuously.

We shall use the modular

$$\rho_{p(\cdot)}(z) := \int_0^T \|z(t)\|_X^{p(t)} dt.$$

Recall that, if  $\|z\|_{L^{p(\cdot)}} \geq 1$ , then

$$\|z\|_{L^{p(\cdot)}}^{p_-} \leq \rho_{p(\cdot)}(z) \leq \|z\|_{L^{p(\cdot)}}^{p_+},$$

whereas the exponents  $p_-$  and  $p_+$  are interchanged when  $\|z\|_{L^{p(\cdot)}} \leq 1$ .

## 5.2. The main Hartman-type theorem

We now introduce the remaining assumptions on the decomposition  $f = m + k$  and state the main result of this chapter.

**(F2<sub>m</sub>)** For a.e.  $t \in [0, T]$  and every  $x, y \in X$ ,

$$\langle m(t, x) - m(t, y), x - y \rangle \geq 0.$$

**(F2<sub>ws</sub>)** For a.e.  $t \in [0, T]$ , whenever  $x_j \rightharpoonup x$  in  $X$ ,

$$k(t, x_j) \longrightarrow k(t, x) \quad \text{in } X^*.$$

**(F3<sub>e</sub>)** There exists  $R > 0$  such that

$$\langle m(t, x) + k(t, x), x \rangle \geq 0$$

for a.e.  $t \in [0, T]$  and every  $x \in X$  satisfying  $\|x\|_X \geq R$ .

The main result of this chapter is the following.

**Theorem 5.2.1** *Assume that  $S$  satisfies (H0)–(H3) and that  $m$  and  $k$  satisfy (F1), (F2<sub>m</sub>), (F2<sub>ws</sub>), and (F3<sub>e</sub>). Then problem (5.1) admits at least one solution.*

The remainder of this section is devoted to the proof of Theorem 5.2.1. Since the perturbation  $k$  may destroy the monotonicity of the full variational operator, a direct Browder–Minty argument is unavailable. We therefore construct finite-dimensional solutions through the projected Lyapunov–Schmidt system and topological degree. In the passage to the limit, monotonicity is used for  $m$ , while weak-to-strong continuity is used only for  $k$ .

We begin by introducing the Galerkin spaces and the corresponding projected principal operators. These finite-dimensional objects preserve the monotonicity and coercivity properties of  $S$  needed in the construction.

If  $X$  is finite-dimensional, the projected argument can be carried out directly in  $X$ . We therefore consider the infinite-dimensional case. Since  $X$  is separable, there exists a linearly independent sequence  $(e_j)_{j \geq 1} \subset X$  such that

$$\overline{\text{span}\{e_j : j \geq 1\}}^X = X.$$

When  $X$  is a Hilbert space, we choose this sequence to be orthonormal with respect to the inner product of  $X$ .

For each  $N \in \mathbb{N}$ , define

$$X_N := \text{span}\{e_1, \dots, e_N\}.$$

We endow  $X_N$  with the auxiliary inner product  $(\cdot, \cdot)_N$  for which  $(e_1, \dots, e_N)$  is an orthonormal basis, and denote the corresponding norm by  $\|\cdot\|_N$ . In the Hilbert-space case,  $(\cdot, \cdot)_N$  is simply the restriction to  $X_N$  of the inner product of  $X$ .

For  $y \in X^*$ , define

$$R_N y := \sum_{j=1}^N \langle y, e_j \rangle e_j.$$

Equivalently,

$$(R_N y, z)_N = \langle y, z \rangle, \quad z \in X_N.$$

In particular, when  $X$  is Hilbert and is identified with  $X^*$  through the Riesz isomorphism,  $R_N$  is the orthogonal projection onto  $X_N$ .

Finally, set

$$S_N(t, x) := R_N S(t, x), \quad (t, x) \in [0, T] \times X_N.$$

**Lemma 5.2.2** *For every  $N \in \mathbb{N}$  and every  $t \in [0, T]$ , the map  $S_N(t, \cdot) : X_N \rightarrow X_N$  is a homeomorphism. Moreover, if*

$$\Gamma_N(t, \cdot) := (S_N(t, \cdot))^{-1},$$

*then  $\Gamma_N : [0, T] \times X_N \rightarrow X_N$  is continuous,  $\Gamma_N(t, \cdot)$  is strictly monotone for every  $t \in [0, T]$ , and*

$$\|\Gamma_N(t, y)\|_N \longrightarrow \infty \quad \text{as } \|y\|_N \rightarrow \infty,$$

uniformly with respect to  $t \in [0, T]$ .

PROOF. For  $x, y \in X_N$ , the definition of  $R_N$  gives

$$(S_N(t, x) - S_N(t, y), x - y)_N = \langle S(t, x) - S(t, y), x - y \rangle.$$

Hence  $S_N(t, \cdot)$  is strictly monotone. Moreover, by **(H2)** and the equivalence of the norms on  $X_N$ ,

$$\frac{(S_N(t, x), x)_N}{\|x\|_N} \longrightarrow \infty \quad \text{as } \|x\|_N \rightarrow \infty,$$

uniformly with respect to  $t \in [0, T]$ . Thus  $S_N(t, \cdot)$  is continuous, coercive, and strictly monotone. The finite-dimensional Browder–Minty theorem implies that it is bijective.

The strict monotonicity of  $\Gamma_N(t, \cdot)$  follows directly from that of  $S_N(t, \cdot)$ . To prove joint continuity, let

$$(t_j, y_j) \rightarrow (t, y), \quad x_j := \Gamma_N(t_j, y_j).$$

Coercivity implies that  $(x_j)$  is bounded. Since  $X_N$  is finite-dimensional, every subsequence has a further subsequence converging to some  $x \in X_N$ . By continuity,

$$S_N(t, x) = y,$$

and therefore  $x = \Gamma_N(t, y)$ . Uniqueness of the limit gives

$$x_j \rightarrow \Gamma_N(t, y).$$

Finally, suppose that  $\|y_j\|_N \rightarrow \infty$  while  $\|\Gamma_N(t_j, y_j)\|_N$  remains bounded. After passing to a subsequence,  $\Gamma_N(t_j, y_j)$  converges in  $X_N$ . The continuity of  $S_N$  would then imply that

$$y_j = S_N(t_j, \Gamma_N(t_j, y_j))$$

is bounded, which is a contradiction. □

Once the projected principal operators have been introduced, we formulate the finite-dimensional counterpart of the generalized Lyapunov–Schmidt system and isolate the topological argument used to solve it.

For  $h_N \in L^1([0, T]; X_N)$ , define

$$H_N(h_N)(t) := \int_0^t h_N(s) ds.$$

As in the construction of the nonlinear Green operator in the preceding chapters, for every  $g_N \in$

$C([0, T]; X_N)$  there exists a unique element  $\widehat{c}_N(g_N) \in X_N$  such that

$$\int_0^T \Gamma_N(t, \widehat{c}_N(g_N) + g_N(t)) dt = 0. \quad (5.2)$$

Moreover,

$$\widehat{c}_N : C([0, T]; X_N) \rightarrow X_N$$

is continuous and maps bounded sets into bounded sets. This follows from the same argument used in Lemma 3.2.1, with  $\Gamma$  replaced by  $\Gamma_N$ .

Define

$$C_0([0, T]; X_N) := \{v \in C([0, T]; X_N) : v(0) = v(T) = 0\}$$

and the projected nonlinear Green operator

$$\mathcal{T}_N : L^1([0, T]; X_N) \rightarrow C_0([0, T]; X_N)$$

by

$$\mathcal{T}_N(h_N)(t) := \int_0^t \Gamma_N(s, \widehat{c}_N(H_N(h_N)) + H_N(h_N)(s)) ds. \quad (5.3)$$

The operator  $\mathcal{T}_N$  is well defined and continuous.

Let

$$E_N := C_0([0, T]; X_N), \quad \|v\|_{E_N} := \max_{t \in [0, T]} \|v(t)\|_X.$$

For  $\lambda \in [0, 1]$ , consider

$$\begin{cases} \frac{d}{dt} S_N(t, u'_N(t)) = \lambda R_N f(t, u_N(t)), \\ u_N(0) = u_N(T), \quad u'_N(0) = u'_N(T). \end{cases} \quad (5.4)$$

Writing

$$u_N = a_N + v_N, \quad a_N := u_N(0) \in X_N, \quad v_N := u_N - u_N(0) \in E_N,$$

we obtain, for  $\lambda \in (0, 1]$ , the equivalent projected Lyapunov–Schmidt system

$$\begin{cases} v_N = \mathcal{T}_N(\lambda R_N N_f(a_N + v_N)), \\ R_N Q N_f(a_N + v_N) = 0, \end{cases} \quad (5.5)$$

where  $N_f$  and  $Q$  are defined as in the preceding chapters.

We now give a general existence result for systems similar to the Lyapunov–Schmidt system described above.

**Theorem 5.2.3** *For each  $N \in \mathbb{N}$ , let  $\mathcal{G}_N : [0, 1] \times C([0, T]; X_N) \rightarrow E_N$  and  $Q_N : C([0, T]; X_N) \rightarrow$*

$X_N$  be continuous operators satisfying:

1.  $\mathcal{Q}_N$  maps bounded sets into bounded sets.
2.  $\mathcal{G}_N(0, u) = 0$  for every  $u \in C([0, T]; X_N)$ , and  $\mathcal{G}_N$  maps bounded sets into relatively compact subsets of  $E_N$ .
3. There exists  $C > 0$ , independent of  $N$  and  $\lambda$ , such that every solution of

$$\begin{cases} v = \mathcal{G}_N(\lambda, a + v), \\ \mathcal{Q}_N(a + v) = 0, \end{cases} \quad a \in X_N, \quad v \in E_N, \quad \lambda \in (0, 1],$$

satisfies

$$\|a\|_X + \|v\|_{E_N} \leq C.$$

4. There exist  $\rho > C$  and  $N_0 \in \mathbb{N}$  such that, for every  $N \geq N_0$ ,

$$d_B[\mathcal{Q}_N, B_\rho^{X_N}, 0] \neq 0,$$

where  $B_\rho^{X_N} := \{a \in X_N : \|a\|_X < \rho\}$  (here  $\mathcal{Q}_N$  is restricted to  $X_N$ , since it is a subspace of  $C([0, T]; X_N)$ ).

Then, for every  $N \geq N_0$ , there exist  $a_N \in X_N$  and  $v_N \in E_N$  such that

$$\begin{cases} v_N = \mathcal{G}_N(1, a_N + v_N), \\ \mathcal{Q}_N(a_N + v_N) = 0. \end{cases} \quad (5.6)$$

PROOF. Fix  $N \geq N_0$  and define

$$\mathcal{K}_{N,\lambda}(a, v) := (a - \mathcal{Q}_N(a + v), \mathcal{G}_N(\lambda, a + v)).$$

The fixed points of  $\mathcal{K}_{N,\lambda}$  are precisely the solutions of the corresponding system. Since  $X_N$  is finite-dimensional, assumptions (1) and (2) imply that  $\mathcal{K}_{N,\lambda}$  is completely continuous.

Choose  $M > C$  and set

$$\Omega_N := \{(a, v) \in X_N \times E_N : \|a\|_X < \rho, \quad \|v\|_{E_N} < M\}.$$

By assumption (3), no fixed point lies on  $\partial\Omega_N$  for  $\lambda \in (0, 1]$ . At  $\lambda = 0$ ,

$$(I - \mathcal{K}_{N,0})(a, v) = (\mathcal{Q}_N(a + v), v).$$

Consider the homotopy

$$H_s(a, v) := (\mathcal{Q}_N(a + sv), v), \quad 0 \leq s \leq 1.$$

Every zero of  $H_s$  satisfies  $v = 0$ . Hence a zero on  $\partial\Omega_N$  would satisfy  $\|a\|_X = \rho$  and  $Q_N(a) = 0$ , contrary to assumption (4), since there are no solutions on  $\partial B_\rho^{X_N}$  for  $Q_N(a) = 0$ . By homotopy invariance and the product property of the degree,

$$d_{LS}[I - \mathcal{K}_{N,1}, \Omega_N, 0] = d_{LS}[I - \mathcal{K}_{N,0}, \Omega_N, 0] = d_B[Q_N, B_\rho^{X_N}, 0] \neq 0.$$

Therefore  $\mathcal{K}_{N,1}$  has a fixed point in  $\Omega_N$ .  $\square$

We now apply the abstract degree result to a regularized family of projected problems. The regularization makes the exterior Hartman inequality strict and yields estimates independent of  $N$ .

For each  $N \in \mathbb{N}$ , set

$$\varepsilon_N := \frac{1}{N}$$

and define

$$\tilde{f}_N(t, x) := f(t, x) + \varepsilon_N S(t, x).$$

Let

$$\mathcal{G}_N(\lambda, u) := \mathcal{T}_N \left( \lambda R_N N_{\tilde{f}_N}(u) \right)$$

and

$$Q_N(u) := R_N \int_0^T \tilde{f}_N(t, u(t)) dt.$$

**Proposition 5.2.4** *For every  $N \in \mathbb{N}$ , the regularized projected problem*

$$\begin{cases} \frac{d}{dt} S_N(t, u'_N(t)) = R_N f(t, u_N(t)) + \varepsilon_N R_N S(t, u_N(t)), \\ u_N(0) = u_N(T), \quad u'_N(0) = u'_N(T), \end{cases} \quad (5.7)$$

*admits at least one solution. Moreover, there exists  $C > 0$ , independent of  $N$ , such that*

$$\|u_N\|_{C([0,T];X)} + \|u'_N\|_{L^{p(\cdot)}([0,T];X)} \leq C. \quad (5.8)$$

**PROOF.** The maps  $\mathcal{G}_N$  and  $Q_N$  are continuous. Moreover,  $Q_N$  maps bounded sets into bounded sets, while  $\mathcal{G}_N$  maps bounded sets into relatively compact subsets of  $E_N$ . For each fixed  $N$ , these properties follow from **(F1)**, **(H3)**, the continuity of  $\mathcal{T}_N$ , and the finite-dimensionality of  $X_N$ . Relative compactness follows from the same equicontinuity and uniform boundedness arguments used in Lemma 4.2.1.

Let  $u_N = a_N + v_N$  solve the auxiliary one-parameter problem, with  $\lambda \in (0, 1]$ ,

$$\begin{cases} \frac{d}{dt} S_N(t, u'_N(t)) = \lambda [R_N f(t, u_N(t)) + \varepsilon_N R_N S(t, u_N(t))], \\ u_N(0) = u_N(T), \quad u'_N(0) = u'_N(T), \end{cases} \quad (5.9)$$

Testing the corresponding projected equation with  $u_N$ , integrating by parts, and using the periodicity conditions, we obtain

$$\begin{aligned} c_0 \rho_{p(\cdot)}(u'_N) &= c_0 \int_0^T \|u'_N(t)\|^{p(t)} dt \leq \int_0^T \langle S(t, u'_N(t)), u'_N(t) \rangle dt \\ &\leq -\lambda \int_0^T \langle \tilde{f}_N(t, u_N(t)), u_N(t) \rangle dt. \end{aligned}$$

There exists  $\tau_N \in [0, T]$  such that

$$\|u_N(\tau_N)\|_X < R.$$

Otherwise, **(F3<sub>e</sub>)** and **(H2)** would give

$$\langle \tilde{f}_N(t, u_N(t)), u_N(t) \rangle \geq \varepsilon_N c_0 \|u_N(t)\|_X^{p(t)} > 0$$

for a.e.  $t$ , which contradicts the preceding inequality.

Set

$$O_N := \{t \in [0, T] : \|u_N(t)\|_X < R\}.$$

On  $[0, T] \setminus O_N$ , we have,

$$-\langle \tilde{f}_N(t, u_N(t)), u_N(t) \rangle \leq 0.$$

On  $O_N$ , condition **(F1)** gives

$$-\langle f(t, u_N(t)), u_N(t) \rangle \leq R \eta_R(t), \quad -\varepsilon_N \langle S(t, u_N(t)), u_N(t) \rangle \leq 0$$

Consequently,

$$c_0 \rho_{p(\cdot)}(u'_N) \leq -\lambda \int_0^T \langle \tilde{f}_N(t, u_N(t)), u_N(t) \rangle dt \leq R \|\eta_R\|_{L^1([0, T])}.$$

The modular-norm inequalities imply

$$\|u'_N\|_{L^{p(\cdot)}([0, T]; X)} \leq C$$

with  $C$  independent of  $N$  and  $\lambda$ .

For every  $t \in [0, T]$ ,

$$u_N(t) = u_N(\tau_N) + \int_{\tau_N}^t u'_N(s) ds.$$

Using  $L^{p(\cdot)}([0, T]; X) \hookrightarrow L^1([0, T]; X)$ , we obtain for some constant  $C' > 0$

$$\|u_N(t)\|_X \leq R + \int_0^T \|u'_N(s)\|_X ds \leq R + C \|u'_N\|_{L^{p(\cdot)}([0, T]; X)} \leq C'.$$

Thus, using  $a_N = u_N(0)$  and  $v_N(t) = u_N(t) - u_N(0)$ , we get for some constant  $C_1 > 0$

$$\|a_N\|_X + \|v_N\|_{E_N} \leq C_1$$

uniformly in  $N$  and  $\lambda$ , which proves assumption (3) of Theorem 5.2.3.

Choose

$$\rho > \max\{C_1, R\}.$$

For every  $a \in X_N$  with  $\|a\|_X = \rho$ ,

$$(Q_N(a), a)_N = \int_0^T \langle f(t, a), a \rangle dt + \varepsilon_N \int_0^T \langle S(t, a), a \rangle dt > 0.$$

Hence

$$(1 - \theta)Q_N(a) + \theta a, \quad 0 \leq \theta \leq 1,$$

does not vanish on  $\partial B_\rho^{X_N}$ , and therefore

$$d_B[Q_N, B_\rho^{X_N}, 0] = 1.$$

Theorem 5.2.3 gives a solution of (5.7), and the preceding estimates give (5.8).  $\square$

It remains to pass to the limit in the projected solutions. We first record the density of the Galerkin spaces in the variational space and then complete the proof of the main theorem.

For each  $N \in \mathbb{N}$ , define

$$V_N := \{u \in W^{1, p(\cdot)}([0, T]; X_N) : u(0) = u(T)\}.$$

**Lemma 5.2.5** *The set*

$$\mathcal{V} := \bigcup_{N \in \mathbb{N}} V_N$$

*is dense in  $V$ .*

**PROOF.** Let  $u \in V$  and  $\varepsilon > 0$ . Since  $\bigcup_{N \in \mathbb{N}} X_N$  is dense in  $X$ , there exist  $N_0 \in \mathbb{N}$ ,  $a_{N_0} \in X_{N_0}$ , and a step function  $\sigma : [0, T] \rightarrow X_{N_0}$  such that

$$\|u(0) - a_{N_0}\|_X + \|u' - \sigma\|_{L^{p(\cdot)}([0, T]; X)} < \varepsilon.$$



Set

$$w(t) := a_{N_0} + \int_0^t \sigma(s) ds, \quad \psi(t) := \frac{t}{T}(w(T) - w(0)).$$

Then  $\tilde{w} := w - \psi$  belongs to  $V_{N_0}$ . Since  $u(0) = u(T)$ ,

$$w(T) - w(0) = \int_0^T (\sigma(s) - u'(s)) ds.$$

Using  $L^{p(\cdot)}([0, T]; X) \hookrightarrow L^1([0, T]; X)$ , we obtain

$$\|u - \tilde{w}\|_{W^{1,p(\cdot)}([0,T];X)} \leq C \left( \|u(0) - a_{N_0}\|_X + \|u' - \sigma\|_{L^{p(\cdot)}([0,T];X)} \right).$$

The result follows by choosing the approximations sufficiently close.  $\square$

With the results shown above, we can finally prove the main result of this chapter.

**PROOF OF THEOREM 5.2.1.** Let  $u_N \in V_N$  be the solutions provided by Proposition 5.2.4. By (5.8), after passing to a subsequence,

$$u_N \rightharpoonup u \quad \text{in } V$$

for some  $u \in V$ . Since evaluation at each  $t$  is a continuous linear map from  $V$  into  $X$ ,

$$u_N(t) \rightharpoonup u(t) \quad \text{in } X.$$

Condition **(F2)<sub>wS</sub>** gives

$$k(t, u_N(t)) \longrightarrow k(t, u(t)) \quad \text{in } X^*$$

for a.e.  $t \in [0, T]$ . The uniform  $C([0, T]; X)$  bound and **(F1)** allow us to use dominated convergence in the variable-exponent modular, yielding

$$k(\cdot, u_N(\cdot)) \longrightarrow k(\cdot, u(\cdot)) \quad \text{in } L^{p'(\cdot)}([0, T]; X^*). \quad (5.10)$$

Furthermore, **(H3)** and the same uniform bound imply

$$\varepsilon_N S(\cdot, u_N(\cdot)) \longrightarrow 0 \quad \text{in } L^{p'(\cdot)}([0, T]; X^*). \quad (5.11)$$

Let  $w \in V$ . Since the spaces  $V_N$  are increasing and  $\bigcup_N V_N$  is dense in  $V$ , choose  $w_N \in V_N$  such that

$$w_N \longrightarrow w \quad \text{in } V.$$

In particular,

$$w_N \longrightarrow w \quad \text{in } C([0, T]; X).$$

The Carathéodory property of  $m$  and **(F1)** give

$$m(\cdot, w_N(\cdot)) \longrightarrow m(\cdot, w(\cdot)) \quad \text{in } L^{p'(\cdot)}([0, T]; X^*). \quad (5.12)$$

Taking  $u_N - w_N$  as a test function in (5.7) gives us

$$\begin{aligned} 0 &= \int_0^T \langle S(t, u'_N), u'_N - w'_N \rangle dt + \int_0^T \langle m(t, u_N), u_N - w_N \rangle dt \\ &\quad + \int_0^T \langle k(t, u_N), u_N - w_N \rangle dt + \varepsilon_N \int_0^T \langle S(t, u_N), u_N - w_N \rangle dt. \end{aligned}$$

but since  $S$  and  $m$  are monotone for a.e.  $t \in [0, T]$ , it follows that:

$$\begin{aligned} \int_0^T \langle S(t, u'_N), u'_N - w'_N \rangle dt &\geq \int_0^T \langle S(t, w'_N), u'_N - w'_N \rangle dt, \\ \int_0^T \langle m(t, u_N), u_N - w_N \rangle dt &\geq \int_0^T \langle m(t, w_N), u_N - w_N \rangle dt, \end{aligned}$$

and thus,

$$\begin{aligned} 0 &\geq \int_0^T \langle S(t, w'_N), u'_N - w'_N \rangle dt + \int_0^T \langle m(t, w_N), u_N - w_N \rangle dt \\ &\quad + \int_0^T \langle k(t, u_N), u_N - w_N \rangle dt + \varepsilon_N \int_0^T \langle S(t, u_N), u_N - w_N \rangle dt. \end{aligned}$$

The last term tends to zero by (5.11). By Proposition 2.3.5, conditions **(H0)** and **(H3)** imply that the Nemytskii operator

$$N_S : L^{p(\cdot)}([0, T]; X) \longrightarrow L^{p'(\cdot)}([0, T]; X^*)$$

is continuous. Since  $w'_N \rightarrow w'$  in  $L^{p(\cdot)}([0, T]; X)$ , it follows that

$$S(\cdot, w'_N(\cdot)) \longrightarrow S(\cdot, w'(\cdot)) \quad \text{in } L^{p'(\cdot)}([0, T]; X^*).$$

Using this convergence, (5.10), (5.12), and the weak convergence of  $u_N - w_N$ , we may pass to the limit and obtain

$$0 \geq \int_0^T \langle S(t, w'), u' - w' \rangle dt + \int_0^T \langle m(t, w), u - w \rangle dt + \int_0^T \langle k(t, u), u - w \rangle dt.$$

Fix  $z \in V$  and set

$$w_\delta := u - \delta z.$$

Substituting  $w = w_\delta$  and dividing by  $\delta > 0$ , we obtain

$$0 \geq \int_0^T \langle S(t, u' - \delta z'), z' \rangle dt + \int_0^T \langle m(t, u - \delta z), z \rangle dt + \int_0^T \langle k(t, u), z \rangle dt.$$

The continuity of  $N_S$  gives

$$S(\cdot, u' - \delta z') \rightarrow S(\cdot, u') \quad \text{in } L^{p'(\cdot)}.$$

Moreover, the Carathéodory property of  $m$  and **(F1)** give

$$m(\cdot, u - \delta z) \rightarrow m(\cdot, u) \quad \text{in } L^{p'(\cdot)}.$$

Letting  $\delta \rightarrow 0^+$  yields

$$\int_0^T \langle S(t, u'), z' \rangle dt + \int_0^T \langle m(t, u) + k(t, u), z \rangle dt \leq 0.$$

Replacing  $z$  by  $-z$ , we conclude that

$$\int_0^T \langle S(t, u'), z' \rangle dt + \int_0^T \langle f(t, u), z \rangle dt = 0$$

for every  $z \in V$ . Thus  $u$  is a weak solution.

Set

$$q(t) := S(t, u'(t)).$$

Testing first with compactly supported functions gives

$$q' = f(\cdot, u(\cdot))$$

in the distributional sense. Hence

$$q \in W^{1, p'(\cdot)}([0, T]; X^*) \hookrightarrow C([0, T]; X^*),$$

and the continuity of  $\Gamma$  gives

$$u'(t) = \Gamma(t, q(t)) \in C([0, T]; X).$$

Integration by parts and constant test functions yield  $q(T) = q(0)$ . Since  $S(0, \cdot) = S(T, \cdot)$  and both maps are bijective,

$$\Gamma(0, \cdot) = \Gamma(T, \cdot).$$

Therefore,

$$u'(T) = \Gamma(T, q(T)) = \Gamma(0, q(0)) = u'(0).$$

Since  $u \in V$ , also  $u(T) = u(0)$ . Thus  $u$  is a solution of (5.1). □

## 5.3. The limiting cases

The general decomposition  $f = m + k$  contains two natural limiting cases. When  $k \equiv 0$ , the right-hand side is purely monotone and stronger conclusions are available. At the opposite extreme, when  $m \equiv 0$ , the exterior Hartman condition is necessarily degenerate in infinite dimension.

### 5.3.1. The purely monotone case

The purely monotone case is obtained by setting

$$m = f, \quad k \equiv 0.$$

Accordingly, condition **(F1)** is understood with this decomposition. In this setting, the weak-to-strong condition is automatic and the exterior Hartman condition can be replaced by a condition on a single sphere. We assume:

**(F2)** For a.e.  $t \in [0, T]$  and every  $x, y \in X$ ,

$$\langle f(t, x) - f(t, y), x - y \rangle \geq 0.$$

**(F3)** There exists  $R > 0$  such that

$$\langle f(t, x), x \rangle \geq 0$$

for a.e.  $t \in [0, T]$  and every  $x \in X$  with  $\|x\|_X = R$ .

**Lemma 5.3.1** *Assume (F2) and (F3). Then*

$$\langle f(t, x), x \rangle \geq 0 \quad \text{whenever } \|x\|_X \geq R$$

for a.e.  $t \in [0, T]$ .

**PROOF.** Let  $\|x\|_X = r > R$  and set

$$y := \frac{R}{r}x.$$

Then  $\|y\|_X = R$ , and monotonicity gives

$$0 \leq \langle f(t, x) - f(t, y), x - y \rangle = \left(1 - \frac{R}{r}\right) \langle f(t, x) - f(t, y), x \rangle.$$

Since  $1 - R/r > 0$ ,

$$\langle f(t, x), x \rangle \geq \langle f(t, y), x \rangle = \frac{r}{R} \langle f(t, y), y \rangle \geq 0.$$

□

With this we can prove the main result for the purely monotone case.

**Corollary 5.3.2** *Assume that  $S$  satisfies (H0)–(H3) and that  $f$  satisfies (F1), (F2), and (F3). Then problem (5.1) admits at least one solution. Moreover, any two solutions differ by a constant element of  $X$ . If  $f(t, \cdot)$  is strictly monotone for a.e.  $t \in [0, T]$ , then the solution is unique.*

PROOF. Set

$$m := f, \quad k := 0.$$

Then  $(\mathbf{F2}_m)$  and  $(\mathbf{F2}_{ws})$  hold, while Lemma 5.3.1 gives  $(\mathbf{F3}_e)$ . Hence the existence of a solution follows from Theorem 5.2.1.

For the uniqueness, let  $u$  and  $w$  be two weak solutions. Subtracting their weak formulations and testing with  $u - w$ , we obtain

$$0 = \int_0^T \langle S(t, u') - S(t, w'), u' - w' \rangle dt + \int_0^T \langle f(t, u) - f(t, w), u - w \rangle dt.$$

Both integrals are nonnegative by (H1) and (F2). Hence  $u' = w'$  a.e., and  $u - w$  is constant in time. If  $f(t, \cdot)$  is strictly monotone, the second integral forces this constant to vanish.  $\square$

Thus the purely monotone case is stronger in two respects: the Hartman condition needs to be verified only on the sphere, and monotonicity yields uniqueness up to an additive constant.

### 5.3.2. The purely weak-to-strong case

The opposite limiting case corresponds to

$$m \equiv 0, \quad k = f.$$

In an infinite-dimensional space, this case is necessarily degenerate under the exterior Hartman condition.

**Proposition 5.3.3** *Assume that  $X$  is infinite-dimensional and that the full map  $f(t, \cdot) : X \rightarrow X^*$  satisfies  $(\mathbf{F2}_{ws})$  and  $(\mathbf{F3}_e)$ . Then*

$$\langle f(t, x), x \rangle \geq 0 \quad \text{for every } x \in X$$

*for a.e.  $t \in [0, T]$ . Moreover,  $f(t, 0) = 0$  for a.e.  $t$ , every periodic solution of (5.1) is stationary, and  $u \equiv 0$  is always a solution.*

PROOF. Since  $X$  is separable, reflexive, and infinite-dimensional, there exists a sequence  $(e_j)$  in  $X$

such that  $e_j \rightharpoonup 0$  and  $\|e_j\|_X = 1$ . Fix  $x \in X$  and choose

$$L > R + \|x\|_X.$$

Then

$$\|x + Le_j\|_X \geq L - \|x\|_X > R,$$

and therefore

$$\langle f(t, x + Le_j), x + Le_j \rangle \geq 0.$$

Since  $x + Le_j \rightharpoonup x$ , **(F2<sub>ws</sub>)** gives

$$f(t, x + Le_j) \longrightarrow f(t, x) \quad \text{in } X^*.$$

Moreover,

$$\langle f(t, x + Le_j), Le_j \rangle = L \langle f(t, x + Le_j) - f(t, x), e_j \rangle + L \langle f(t, x), e_j \rangle \longrightarrow 0.$$

Passing to the limit yields

$$\langle f(t, x), x \rangle \geq 0.$$

Applying this inequality to  $\varepsilon v$  and  $-\varepsilon v$ , and letting  $\varepsilon \rightarrow 0^+$ , gives

$$\langle f(t, 0), v \rangle = 0 \quad \text{for every } v \in X.$$

Thus  $f(t, 0) = 0$ .

Finally, testing the weak formulation with  $u$  gives

$$0 = \int_0^T \langle S(t, u'), u' \rangle dt + \int_0^T \langle f(t, u), u \rangle dt.$$

Both terms are nonnegative, so **(H2)** implies  $u' = 0$ . Hence every periodic solution is stationary. Since  $S(t, 0) = 0$  and  $f(t, 0) = 0$ , the zero function is a solution.  $\square$

## 5.4. The radial Hilbert-space case

Throughout this section,  $X$  is a Hilbert space and

$$S(t, 0) = 0, \quad S(t, x) = \beta(t, \|x\|_X) \mathcal{R}x, \quad x \neq 0,$$

where  $\mathcal{R} : X \rightarrow X^*$  is the Riesz isomorphism and  $\beta \in C([0, T] \times (0, \infty); (0, \infty))$  with  $\beta(0, r) = \beta(T, r)$ .

The following result extends to the present Hilbert-space setting the pointwise Hartman estimates obtained by Fan and Fan [7] and García-Huidobro, Manásevich, Mawhin, and Tanaka [8].

**Proposition 5.4.1** *Under the assumptions of this section, suppose that the hypotheses of Theorem 5.2.1 or Corollary 5.3.2 hold. Then problem (5.1) admits a solution satisfying*

$$\|u(t)\|_X \leq R, \quad t \in [0, T].$$

PROOF. Let  $(u_N)_N$  be the regularized projected solutions used in the proof of Theorem 5.2.1. We claim that

$$\|u_N(t)\|_X \leq R, \quad t \in [0, T].$$

Suppose, by contradiction, that

$$\max_{t \in [0, T]} \|u_N(t)\|_X > R.$$

Set

$$\varphi_N(t) := \frac{1}{2} \|u_N(t)\|_X^2$$

and choose  $\sigma \in [0, T)$  such that

$$\varphi_N(\sigma) = \max_{t \in [0, T]} \varphi_N(t).$$

Then

$$\varphi'_N(\sigma) = 0.$$

Indeed, this is immediate if  $\sigma \in (0, T)$ . If  $\sigma = 0$ , maximality gives

$$\varphi'_N(0) \leq 0, \quad \varphi'_N(T) \geq 0,$$

while the periodicity of  $u_N$  and  $u'_N$  implies

$$\varphi'_N(0) = \varphi'_N(T).$$

The radial form of  $S_N$  gives

$$(S_N(\sigma, u'_N(\sigma)), u_N(\sigma))_X = 0.$$

Indeed, this is immediate if  $u'_N(\sigma) = 0$ . Otherwise,

$$\begin{aligned} (S_N(\sigma, u'_N(\sigma)), u_N(\sigma))_X &= \beta(\sigma, \|u'_N(\sigma)\|_X) (u'_N(\sigma), u_N(\sigma))_X \\ &= \beta(\sigma, \|u'_N(\sigma)\|_X) \varphi'_N(\sigma) = 0. \end{aligned}$$

Since  $\sigma < T$  and  $\|u_N(\sigma)\|_X > R$ , there exists  $\delta > 0$  such that  $\sigma + \delta \leq T$  and

$$\|u_N(t)\|_X > R, \quad t \in [\sigma, \sigma + \delta].$$

For a.e.  $t$  in this interval, the projected equation gives

$$\begin{aligned} \frac{d}{dt} (S_N(t, u'_N(t)), u_N(t))_X &= (R_N f(t, u_N(t)) + \varepsilon_N R_N S(t, u_N(t)), u_N(t))_X \\ &\quad + (S_N(t, u'_N(t)), u'_N(t))_X. \end{aligned}$$

Since  $u_N(t) \in X_N$ ,

$$\begin{aligned} &(R_N f(t, u_N(t)) + \varepsilon_N R_N S(t, u_N(t)), u_N(t))_X \\ &= \langle f(t, u_N(t)), u_N(t) \rangle + \varepsilon_N \langle S(t, u_N(t)), u_N(t) \rangle > 0. \end{aligned}$$

Moreover,

$$(S_N(t, u'_N(t)), u'_N(t))_X = \begin{cases} \beta(t, \|u'_N(t)\|_X) \|u'_N(t)\|_X^2, & u'_N(t) \neq 0, \\ 0, & u'_N(t) = 0, \end{cases}$$

so this term is nonnegative. Therefore,

$$\frac{d}{dt} (S_N(t, u'_N(t)), u_N(t))_X > 0$$

for a.e.  $t \in [\sigma, \sigma + \delta]$ . Since the pairing vanishes at  $t = \sigma$ , it follows that

$$(S_N(t, u'_N(t)), u_N(t))_X > 0$$

for every  $t \in (\sigma, \sigma + \delta]$ .

In particular,  $u'_N(t) \neq 0$  on this interval. Hence

$$\begin{aligned} (S_N(t, u'_N(t)), u_N(t))_X &= \beta(t, \|u'_N(t)\|_X) (u'_N(t), u_N(t))_X \\ &= \beta(t, \|u'_N(t)\|_X) \varphi'_N(t). \end{aligned}$$

Since  $\beta > 0$ , we obtain

$$\varphi'_N(t) > 0, \quad t \in (\sigma, \sigma + \delta],$$

contradicting the maximality of  $\varphi_N(\sigma)$ . Thus

$$\|u_N(t)\|_X \leq R, \quad t \in [0, T].$$

Finally, the solution of the original problem is obtained as a weak limit in  $V$  of a subsequence of



$(u_N)$ . Since evaluation at each  $t \in [0, T]$  is continuous from  $V$  into  $X$ ,

$$u_N(t) \rightharpoonup u(t) \quad \text{in } X.$$

The closed ball of radius  $R$  is weakly closed, and therefore

$$\|u(t)\|_X \leq R, \quad t \in [0, T].$$

□

## 5.5. Application to a spatially nonlocal periodic PDE

Let  $\Omega \subset \mathbb{R}^d$  be a bounded domain with Lipschitz boundary. Let  $p \in C([0, T])$  satisfy

$$1 < p_- \leq p(t) \leq p_+ < \infty, \quad p(0) = p(T),$$

and let  $b \in C([0, T]; H^{-1}(\Omega))$  be nonconstant.

We consider the periodic Sobolev-type problem

$$\begin{cases} \frac{\partial}{\partial t} \left[ -\|\nabla u_t(t, \cdot)\|_{L^2(\Omega)}^{p(t)-2} \Delta u_t \right] = -\frac{\Delta u}{\sqrt{1 + \|\nabla u(t, \cdot)\|_{L^2(\Omega)}^2}} + b(t) & \text{in } H^{-1}(\Omega), \\ u(t, \xi) = 0 & (t, \xi) \in [0, T] \times \partial\Omega, \\ u(0, \xi) = u(T, \xi), \quad u_t(0, \xi) = u_t(T, \xi) & \xi \in \Omega. \end{cases} \quad (5.13)$$

**Proposition 5.5.1** *Assume that*

$$B := \|b\|_{C([0, T]; H^{-1}(\Omega))} < 1.$$

*Then problem (5.13) admits a unique nonstationary periodic solution  $u \in C^1([0, T]; H_0^1(\Omega))$  satisfying*

$$\|\nabla u(t, \cdot)\|_{L^2(\Omega)} \leq \frac{B}{\sqrt{1 - B^2}}, \quad t \in [0, T].$$

**PROOF.** Set

$$X := H_0^1(\Omega), \quad X^* := H^{-1}(\Omega), \quad \|x\|_X := \|\nabla x\|_{L^2(\Omega)}.$$

Let  $A : X \rightarrow X^*$  be the Riesz isomorphism defined by

$$\langle A(x), z \rangle := \int_{\Omega} \nabla x \cdot \nabla z \, d\xi.$$

Thus

$$\|A(x)\|_{X^*} = \|x\|_X, \quad A(x) = -\Delta x$$

in the distributional sense.

Define

$$S(t, 0) := 0, \quad S(t, x) := \|x\|_X^{p(t)-2} A(x), \quad x \neq 0,$$

and

$$f(t, x) := \frac{A(x)}{\sqrt{1 + \|x\|_X^2}} + b(t).$$

With this notation, problem (5.13) has the abstract form considered in this chapter.

Since  $p(0) = p(T)$ , **(H0)** holds. Moreover, the standard Hilbert-space inequalities for power-type maps imply that, for every  $M > 0$ , there exists  $c_M > 0$  such that

$$\langle S(t, x) - S(t, y), x - y \rangle \geq c_M \min \{ \|x - y\|_X^2, \|x - y\|_X^{p_+} \}$$

whenever  $\|x\|_X, \|y\|_X \leq M$ . Thus **(H1)** holds. Furthermore,

$$\langle S(t, x), x \rangle = \|x\|_X^{p(t)}, \quad \|S(t, x)\|_{X^*} = \|x\|_X^{p(t)-1},$$

so  $S$  satisfies **(H0)–(H3)**. Notice also that  $S$  has the radial form

$$S(t, x) = \beta(t, \|x\|_X) A(x), \quad \beta(t, r) := r^{p(t)-2}.$$

The functional

$$\Psi(x) := \sqrt{1 + \|x\|_X^2}$$

is strictly convex and satisfies

$$D\Psi(x) = \frac{A(x)}{\sqrt{1 + \|x\|_X^2}}.$$

Consequently,  $f(t, \cdot)$  is strictly monotone. Moreover,

$$\|f(t, x)\|_{X^*} \leq 1 + \|b(t)\|_{X^*},$$

and hence **(F1)** holds.

Set

$$R_B := \frac{B}{\sqrt{1 - B^2}}.$$

For every  $t \in [0, T]$  and  $x \in X$ ,

$$\langle f(t, x), x \rangle = \frac{\|x\|_X^2}{\sqrt{1 + \|x\|_X^2}} + \langle b(t), x \rangle \geq \|x\|_X \left( \frac{\|x\|_X}{\sqrt{1 + \|x\|_X^2}} - B \right).$$

If  $\|x\|_X = R_B$ , then

$$\frac{\|x\|_X}{\sqrt{1 + \|x\|_X^2}} = B,$$

so **(F3)** holds with  $R = R_B$ . Corollary 5.3.2, strict monotonicity, and Proposition 5.4.1 yield existence, uniqueness, and the estimate

$$\|u(t)\|_X \leq \frac{B}{\sqrt{1 - B^2}}, \quad t \in [0, T].$$

Finally, if  $u(t) \equiv c$  were stationary, then

$$\frac{A(c)}{\sqrt{1 + \|c\|_X^2}} + b(t) = 0$$

for every  $t \in [0, T]$ , which would force  $b$  to be constant. Therefore, the unique periodic solution is nonstationary.  $\square$

**Remark 5.5.2** (Weak-to-strong perturbations) Let  $k : [0, T] \times X \rightarrow X^*$ , with  $k \not\equiv 0$ , satisfy the local boundedness condition **(F1)** and **(F2<sub>ws</sub>)**, and suppose that there exist  $\alpha, C \geq 0$  such that

$$\langle k(t, x), x \rangle \geq -\alpha \|x\|_X - C,$$

for a.e.  $t \in [0, T]$  and every  $x \in X$ , with  $\alpha + \|b\|_{C([0, T]; X^*)} < 1$ .

Consider

$$\tilde{f}(t, x) := \frac{A(x)}{\sqrt{1 + \|x\|_X^2}} + k(t, x) + b(t).$$

Since

$$\frac{\|x\|_X^2}{\sqrt{1 + \|x\|_X^2}} \geq \|x\|_X - 1,$$

we obtain

$$\langle \tilde{f}(t, x), x \rangle \geq (1 - \alpha - \|b\|_{C([0, T]; X^*)}) \|x\|_X - (1 + C).$$

Hence **(F3<sub>e</sub>)** holds for a sufficiently large radius. Therefore, Proposition 5.4.1 yield a periodic solution satisfying

$$\|u(t)\|_X \leq R, \quad t \in [0, T].$$

# Chapter 6

## Conclusion

In this thesis we have developed an abstract framework for the study of periodic solutions of nonlinear differential equations in Banach spaces driven by generalized variable-exponent operators. The motivation was to understand to what extent techniques that are classical in finite-dimensional periodic problems can be adapted to a setting where compactness is no longer automatic and where the principal operator is defined between a Banach space and its dual.

The results obtained here show that such an extension is possible, but only after reformulating the problem in a way that separates the different roles played by the periodic boundary conditions. The generalized Lyapunov–Schmidt system derived in Chapter 3 separates the fixed point equation from the averaged constraint imposed by periodicity. This reformulation is used in the mean-admissible and Galerkin approaches.

The first approach, based on what we called *mean-admissible* nonlinearities, shows that the averaged constraint can sometimes be solved directly with respect to the constant component of the solution. In this case, the original periodic problem is reduced to a fixed point problem. The main difficulty is then the lack of compactness in Banach spaces, which is overcome by working within a compact Gelfand triple. This leads to an existence theorem based on Leray–Schauder degree and a priori bounds.

The second approach is based on a Hartman-type condition for nonlinearities of the form  $f(t, x) = m(t, x) + k(t, x)$ , where  $m$  is monotone and  $k$  is sequentially weak-to-strong continuous. Galerkin approximations, topological degree, and a Minty argument yield a periodic solution.

The purely monotone case follows by taking  $k \equiv 0$ . In this setting, the Hartman condition need only be imposed on a sphere and additional uniqueness properties are obtained. By contrast, the purely weak-to-strong case is degenerate in infinite-dimensional spaces. For radial operators in Hilbert spaces, the solution also satisfies the corresponding pointwise Hartman bound.

The methods developed in the thesis are therefore complementary. The mean-admissible ap-

proach solves the averaged equation and recovers compactness through a Gelfand triple, whereas the Hartman approach combines Galerkin approximation, topological degree, monotonicity, and weak-to-strong continuity.

To the best of our knowledge, these results are new and extend the frameworks of [8, 17] to the Banach-space setting considered here.

Several questions remain open. A natural continuation of this thesis is to study nonlinearities depending also on the derivative, namely problems of the form

$$\frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t)).$$

Such problems would require a more careful treatment of the term involving  $u'$ , since the estimates used in this thesis depend strongly on the fact that the nonlinearity acts only on  $u$ . In particular, one would need conditions ensuring that the derivative dependence does not destroy the compactness or monotonicity properties used in the existence arguments.

Another natural direction is to go beyond existence and study bifurcation phenomena for this class of periodic problems. In the present thesis, the main focus was to obtain at least one solution under suitable structural assumptions. A further step would be to understand how solutions behave when parameters are introduced in the equation, and whether new solutions may appear from known or trivial ones. This is not immediate in the present framework, since the principal operator is nonlinear and does not fit directly into the classical Fredholm setting usually used in bifurcation theory.

Finally, the results suggest that the framework developed in this thesis may be useful beyond the abstract setting considered here. In particular, the combination of topological degree, monotonicity, compactness, and Galerkin approximation provides a basis for studying other periodic problems in Banach spaces driven by nonlinear operators with nonstandard growth.

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